# Revisiting Adaptively Secure IBE from Lattices with Smaller Modulus: A Conceptually Simple Framework with Low Overhead†

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**Abstract.** Most adaptively secure identity-based encryption (IBE) constructions from lattices in the standard model follow the framework proposed by Agrawal et al. (EUROCRYPT 2010). However, this framework has an inherent restriction: the modulus is quadratic in the trapdoor norm. This leads to an unnecessarily large modulus, reducing the efficiency of the IBE scheme.

In this paper, we propose a novel framework for adaptively secure lattice-based IBE in the standard model, that removes this quadratic restriction of modulus while keeping the dimensions of the master public key, secret keys, and ciphertexts unchanged. More specifically, our key observation is that the original framework has a *natural* cross-multiplication structure of trapdoor. Building on this observation, we design two novel algorithms with non-spherical Gaussian outputs that fully utilize this structure and thus remove the restriction. Furthermore, we apply our framework to various IBE schemes with different partitioning functions in both integer and ring settings, demonstrating its significant improvements and broad applicability.

Besides, compared to a concurrent[1] work by Ji et al. (PKC 2025), our framework is significantly simpler in design, and enjoys a smaller modulus, a more compact master public key and shorter ciphertexts.

**Keywords:** Lattice-based cryptography · Identity-based encryption · Gaussian convolution.

## 1 Introduction

Identity-based encryption (IBE), proposed by Shamir [Sha84], as a way to simplify public key and certificate management, is a generalization of public key

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[1] This is an updated version of the paper. The original version contained an inaccurate description of the relationship between these two papers. Now the term "independent" has been removed.

encryption, where the public key can be an arbitrary string, such as a name, a telephone number, or an email address. Since its first realization by Boneh and Franklin [BF01], various IBEs based on bilinear maps/pairings [CHK03, BB04a, BB04b, Wat05, Gen06, Wat09], quadratic residuosity [Coc01, BGH07], lattices [GPV08, ABB10, Boy10, CHKP10, DLP14, BL16, KY16, Yam16, ZCZ16, Yam17, LLW20, ALWW21, JKN21, Abl24, JWLG25], and (computational) Diffie Hellman [DG17] have been proposed.

There are two security notions in the literature: selective security and adaptive security. The former requires the adversary to declare the challenge identity before seeing the master public key ($\mathsf{mpk}$). It is usually not strong enough for practical applications. In this paper, we focus on lattice-based IBE constructions with adaptive security in the standard model.

*Adaptively Secure IBE from Lattices in the Standard Model.* In Eurocrypt 2010, Agrawal et al. [ABB10] proposed an efficient adaptively secure IBE in the standard model, avoiding the bit-by-bit IBE construction proposed by Cash et al. [CHKP10]. However, this construction is not compact since the public parameter contains $O(\lambda)$ number of basic matrices, where $\lambda$ is the security parameter.

To deal with this issue, Yamada [Yam16] used the fully key homomorphic technique of Boneh et al. [BGG$^+$14], constructed two IBE schemes from lattices with novel ideas that are different from the Waters' hash [Wat05, ABB10] or the admissible hash [BB04b, CHKP10], and reduced the number of the public matrices to $O(\lambda^{1/d})$ in their first scheme, where $d \in \mathbb{N}$ is an arbitrary constant. However, this scheme requires a super-polynomial LWE modulus. Subsequently, Katsumata and Yamada [KY16] proposed a more efficient IBE scheme from the ring LWE (RLWE) assumption with asymptotically the same number of public matrices, but with only a polynomial modulus. Independently, Zhang et al. [ZCZ16] proposed the first lattice-based programmable hash function, and then obtained an adaptively secure IBE construction whose public parameter contains $O(\log Q)$ number of basic matrices, where $Q$ is the number of the key extraction queries. However, $Q$ must be determined at the setup phase and their scheme only achieves $Q$-bounded security, i.e., the security is not guaranteed anymore if the adversary makes more than $Q$ key extraction queries. Later, Yamada [Yam17] modified the admissible hash and the Waters' hash respectively, and proposed two new constructions with poly-logarithmic public parameter size. The first one achieves $\omega(\log^2 \lambda)$, and the second one achieves $\omega(\log \lambda)$ but relies on the Barrington's Theorem [Bar89] to compute an NC1 boolean circuit, which can be done in polynomial time in theory yet would not be expected to be efficient in practice.

Recently, Jager et al. [JKN21] proposed a new technique called blockwise partitioning and constructed an adaptively secure IBE scheme with the public parameter containing $O(\log \lambda)$ matrices. However, their construction and security analysis relied on the so-called near collision-resistant hash function, and it is not known if this kind of hash function can be constructed from LWE or any variants of LWE. Concurrently, Abla et al. [ALWW21] designed an almost pairwise independent hash and proposed a more compact IBE scheme from the

RLWE assumption with only $\omega(1)$ ring vectors in the public parameter. However, their scheme leverages elegant structures of cyclotomic rings, and it does not appear to be adaptable to the case of plain LWE. Subsequently, Abla [Abl24] used the Lagrange interpolation theorem to design a new equality test function over integers and constructed an adaptively secure IBE scheme that contains $\omega\left(\frac{\log \lambda}{\log \log \lambda}\right)$ public matrices.

*Limitation of the Current Framework.* All the aforementioned IBE schemes follow the framework of [ABB10], differing only in the design of their partitioning functions—for example, (modified) Waters' hash function, (modified) admissible hash function, programmable hash function, near collision-resistant hash function, and almost pairwise independent hash function. However, this framework suffers from an inherent restriction:

*the modulus is quadratic in the trapdoor norm.*

Specifically, this framework necessitates a heavy homomorphic computation of the partitioning function, and the resulting modulus is at least quadratic in the norm of the trapdoor derived from this homomorphic computation. This restriction results in an unnecessarily large modulus (e.g., $\tilde{O}(\lambda^{9.5})$), leading to large total sizes of both the master public key and ciphertexts, along with a long running time of the scheme. This motivates the following question:

*Can we design a novel framework for adaptively secure IBE from lattices, that removes the quadratic restriction of modulus in the trapdoor norm?*

Concurrently, Ji et al. [JWLG25] provide a positive answer to this question. However, their framework has two weaknesses. On the construction side, it is more cumbersome than the original one, requiring extensive modifications including: (1) public matrices in two different encoding forms, (2) extra homomorphic computation for an "incomplete decryption" function, and (3) more complex designs of the user's public key, secret key and the ciphertexts. On the performance side, while successfully removing the quadratic restriction, it introduces extra $\tilde{O}(\lambda^2)$ overhead in the modulus and increases the dimensions of both the master public key and the ciphertexts. Thus, this motivates us to revisit this question and pursue a simpler and better-performing solution.

### 1.1 Our Contribution

In this paper, we revisit the above question and provide a simpler and better-performing solution. In more detail,

– We propose a novel framework for adaptively secure IBE from lattices in the standard model that removes the quadratic restriction of modulus while keeping the dimensions of the master public key, secret key, and ciphertexts unchanged. Specifically, our key observation is that the original framework has a *natural* cross-multiplication structure of trapdoor. Building on this observation, we design two novel algorithms with non-spherical Gaussian outputs that efficiently exploit this structure and thus remove the restriction.

More precisely, our two novel algorithms with non-spherical Gaussian outputs are: (1) a secret key sampling algorithm that generates outputs following a half-large-half-small distribution; (2) an error re-randomization algorithm that generates outputs following a half-small-half-large distribution. These two novel algorithms are of independent interest, and when combined with the natural cross-multiplication structure, could find rich application scenarios in lattice-based cryptography, such as attributed-based encryption and identity-based inner-product functional encryption.

**Table 1.** Modulus size comparison of lattice IBE constructions under three frameworks: (1) original framework, (2) Ji et al.'s framework, (3) our framework.

| Scheme / Framework | [ABB10] + [Boy10]$^\sharp$ | [ZCZ16]$^\star$ | [KY16]$^\natural$ | [Yam17] I + [Kat17]$^*$ | [LLW20]$^\flat$ | [JKN21] | [ALWW21]$^\dagger$ | [Abl24]$^\circ$ |
|---|---|---|---|---|---|---|---|---|
| Original | $\tilde{O}(n^4)$ | $\tilde{O}(n^7Q^2)$ | $\tilde{O}(n^{2d+2})$ | $\tilde{O}(n^6)$ | $\tilde{O}(n^6)^\diamond$ | $\tilde{O}(n^6)$ | $\tilde{O}(n^{10+\frac{4}{\varrho}})$ | $\tilde{O}(n^{10})$ |
| Ji et al.'s$^\ddagger$ | $\tilde{O}(n^{4.5})$ | $\tilde{O}(n^6Q)$ | $\tilde{O}(n^{d+3.5})$ | $\tilde{O}(n^{5.5})$ | $\tilde{O}(n^{5.5})$ | $\tilde{O}(n^{5.5})$ | $\tilde{O}(n^{7.5+\frac{2}{\varrho}})$ | $\tilde{O}(n^{7.5})$ |
| Ours | $\tilde{O}(n^{2.5})$ | $\tilde{O}(n^4Q)$ | $\tilde{O}(n^{d+1.5})$ | $\tilde{O}(n^{3.5})$ | $\tilde{O}(n^{3.5})$ | $\tilde{O}(n^{3.5})$ | $\tilde{O}(n^{5.5+\frac{2}{\varrho}})$ | $\tilde{O}(n^{5.5})$ |

All the schemes set the lattice dimension $n = \Theta(\lambda)$. In this table, we omit the IBE constructions with super-poly modulus [Yam16], [BL16]$^\flat$ or implicit modulus [Yam17], since we focus on showing our improvement in reducing the modulus.

$^\sharp$ Boyen [Boy10] construct block-diagonal matrices with FRD entries and thus removed the restriction $q = 2Q$ in [ABB10].
$^\star$ $Q$ denotes the number of key extraction queries in attacking the IBE scheme.
$^\natural$ $d \in \mathbb{N}$ is a constant typically chosen to be small (e.g., $d = 2$ or 3).
$^*$ Katsumata [Kat17] improved the subset predicate of [Yam17] and thus lowered the approximation factor of the LWE problem from $\tilde{O}(n^{11})$ to $\tilde{O}(n^{5.5})$.
$^\flat$ Boyen-Li [BL16] and Lai et al. [LLW20], constructed adaptively secure IBE schemes with (almost) tight security following the Katz-Wang framework [KW03]. Although the [KW03] framework differs from [ABB10], both require heavy homomorphic computations of a specific function (e.g., pseudorandom function or partitioning function), and share the same restriction that the modulus is quadratic in the trapdoor norm. In this paper, we propose some methods to remove this restriction of [ABB10], which can easily be adapted to the [KW03] framework.
$^\dagger$ $\varrho \geq 1$ is a constant such that $n^{\frac{1}{\varrho}} > 3 + \varrho$, depending on the underlying error correcting code.
$^\circ$ In [Abl24], the modulus was presented as $q = \tilde{O}(n^9)$. However, we re-calculate the modulus and find out that $q = \tilde{O}(n^{10})$. This re-calculation result is confirmed through personal communications with the author and we present our result in the table.
$^\diamond$ Lai et al. [LLW20] homomorphically compute a specific decryption circuit in NC1 and obtain an expanding factor of $O(n^{1.5+\upsilon})$ for any $\upsilon \in (0, 1)$. Here, we set $\upsilon = 0.5$.
$^\ddagger$ Ji et al. [JWLG25] proposed a framework that removes the quadratic restriction, but only applied their framework to a single IBE [ALWW21]. Here, we extend their approach to various IBEs.

– We apply our framework to all aforementioned IBE schemes with different partitioning functions in both integer and ring settings, demonstrating its significant improvements and broad applicability. Please refer to Tab. 1 for a modulus size comparison among our framework, Ji et al.'s framework, and the original framework. For a more detailed performance comparison, please refer to Tab. 2.

Compared to Ji et al.'s framework [JWLG25], our framework achieves not only (asymptotically) a smaller modulus, but also (concretely) a more compact master public key and shorter ciphertexts. Furthermore, our framework is significantly simpler in design. Specifically, they make extensive modifications to the original framework to generate an *artificial* cross-multiplication structure, while we only need to use the *natural* cross-multiplication structure of the original framework. Nevertheless, their framework has its own merits: their secret key size is the smallest compared to both our framework and the original framework. Due to space limitations, we provide a clear review of the techniques by Ji et al. [JWLG25] and a detailed comparison in Appendix A.

### 1.2 Roadmap

In Sect. 2, we outline the target problems and provide an overview of our techniques. In Sect. 3, we provide some preliminary materials, such as lattices, Gaussians, and IBE. Sect. 4 and Sect. 5 introduce our new sampling and noise rerandomization algorithms. In Sect. 6, we present a general framework of lattice IBE, incorporating our new techniques. In Sect. 7, we apply our new framework to various IBEs and show performance improvements.

## 2 Techniques Overview

**Original lattice IBE framework and its quadratic restriction.** We abstract the original framework [ABB10][2] for adaptively secure IBE from lattices in the standard model and revisit its quadratic restriction on the modulus.

In previous works, $\mathsf{mpk}$ mainly consists of $t$[3] BGG+-style encoding [BGG+14]:

$$\mathsf{mpk} \coloneqq \big(\mathbf{B},\ \{\mathbf{C}_i \coloneqq \mathbf{B}\mathbf{R}_i + \kappa_i \mathbf{G}\}_{i\in[t]}\big), \tag{1}$$

where $\mathbf{B} \in \mathbb{Z}_q^{n\times m}$ is generated along with a secret trapdoor $\mathbf{T_B} \in \mathbb{Z}^{m\times m}$ such that $\mathbf{B}\cdot\mathbf{T_B} = \mathbf{0} \bmod q$, $\mathbf{G} \in \mathbb{Z}_q^{n\times m}$ is the public gadget matrix introduced in [MP12] with a well-known trapdoor $\mathbf{T_G}$, $\mathbf{R}_i \in \mathbb{Z}^{m\times m}$ is a random matrix with small entries, $\kappa_i \in \mathbb{Z}$ is the randomness used to define the partitioning function $H(\cdot,\cdot)$[4]. Boneh et al. [BGG+14] showed that the BGG+-style encoding supports homomorphic addition and multiplication operations. Thus, a user's

[2] Here, we need to clarify that "the original framework" is not the framework used in the original [ABB10] paper, but is rather a general framework (completed by [KY16]) that can be applied to [ABB10] and subsequent works.

[3] $t \in \mathbb{N}$ is a parameter determined by the specific partitioning function, e.g., $t = \omega(1)$ in [ALWW21], $t = \omega(\log\lambda)$ in [Yam17], or $t = O(\lambda)$ in [ABB10].

[4] $H(\cdot,\cdot)$ is called "partitioning function" since the event $H(\boldsymbol{\kappa}, \mathsf{id}^{(1)}) \neq 0 \wedge \cdots H(\boldsymbol{\kappa}, \mathsf{id}^{(Q)}) \neq 0 \wedge H(\boldsymbol{\kappa}, \mathsf{id}^*) = 0$ happens with a noticeable probability, where $\boldsymbol{\kappa} \coloneqq (\kappa_1, \cdots, \kappa_t)$.

public key $\mathsf{pk}_{\mathsf{id}}$, which is homomorphically computed from its identity $\mathsf{id}$ and $\mathsf{mpk}$[5], can be written as

$$\mathsf{pk}_{\mathsf{id}} := [\mathbf{B}|\mathbf{B}\mathbf{R}_{\mathsf{id}} + H(\boldsymbol{\kappa}, \mathsf{id})\mathbf{G}] \in \mathbb{Z}_q^{n\times 2m}.$$

For a given uniform vector $\mathbf{u} \in \mathbb{Z}_q^n$, the user's secret key $\mathsf{sk}_{\mathsf{id}}$, which is a $2m$-length vector satisfying $\mathsf{pk}_{\mathsf{id}} \cdot \mathsf{sk}_{\mathsf{id}} = \mathbf{u}$, can be sampled with the secret trapdoor $\mathbf{R}_{\mathsf{id}}$ and the public trapdoor $\mathbf{T}_{\mathbf{G}}$. Specifically, $\mathsf{sk}_{\mathsf{id}}$ satisfies the following equation

$$\underbrace{[\mathbf{B}|\mathbf{B}\mathbf{R}_{\mathsf{id}} + H(\boldsymbol{\kappa}, \mathsf{id})\mathbf{G}]}_{:=\mathsf{pk}_{\mathsf{id}}} \cdot \underbrace{\left(\mathbf{p}_{\mathsf{sk}} + \begin{bmatrix} -\mathbf{R}_{\mathsf{id}} \\ \mathbf{I} \end{bmatrix} \mathbf{z}\right)}_{:=\mathsf{sk}_{\mathsf{id}}} = \mathbf{u}. \tag{2}$$

where $\mathbf{p}_{\mathsf{sk}} \in \mathbb{Z}^{2m}$ is a perturbation to ensure that no information about the trapdoor matrix $\mathbf{R}_{\mathsf{id}}$ is revealed and $\mathbf{z} \in \mathbb{Z}^m$ follows a discrete Gaussian distribution such that $\mathbf{G}\mathbf{z} = H(\boldsymbol{\kappa}, \mathsf{id})^{-1} \cdot (\mathbf{u} - \mathsf{pk}_{\mathsf{id}}\mathbf{p}_{\mathsf{sk}})$. The Gaussian convolution technique [Pei10] guarantees that $\mathsf{sk}_{\mathsf{id}}$ is distributed statistically close to a spherical Gaussian $D_{\mathbb{Z}^{2m},\theta}$ under the condition that $\theta$ is *linear* in $\|\mathbf{R}_{\mathsf{id}}\|$.

The ciphertext of a message $\mu \in \{0, 1\}$ consists of two parts:

$$\mathsf{ct} := \left(c_0 := \mathbf{v}^\top\mathbf{u} + e_0 + \left\lceil \frac{q}{2} \right\rceil \cdot \mu, \quad \mathbf{c}^\top := \mathbf{v}^\top \mathsf{pk}_{\mathsf{id}} + \mathbf{e}^\top\right) \in \mathbb{Z}_q \times \mathbb{Z}_q^{2m},$$

where $\mathbf{v} \in \mathbb{Z}_q^n$ is a secret and $(e_0, \mathbf{e}) \leftarrow (D_{\mathbb{Z},\delta}, D_{\mathbb{Z}^{2m},\sigma})$ are some errors. The public key for the challenge identity $\mathsf{id}^*$ can be written as $\mathsf{pk}_{\mathsf{id}^*} := [\mathbf{B}|\mathbf{B}\mathbf{R}_{\mathsf{id}^*}]$ since $H(\boldsymbol{\kappa}, \mathsf{id}^*) = 0$. Thus, the $2m$-length vector $\mathbf{c}^*$ in the challenge ciphertext can be simulated as follows

$$\mathbf{c}^* := \begin{bmatrix} \mathbf{B}^\top\mathbf{v} + \mathbf{e}_0' \\ \mathbf{R}_{\mathsf{id}^*}^\top(\mathbf{B}^\top\mathbf{v} + \mathbf{e}_0') \end{bmatrix} + \mathbf{p}_{\mathbf{e}} = \begin{bmatrix} \mathbf{B}^\top \\ \mathbf{R}_{\mathsf{id}^*}^\top\mathbf{B}^\top \end{bmatrix} \mathbf{v} + \underbrace{\left(\begin{bmatrix} \mathbf{I} \\ \mathbf{R}_{\mathsf{id}^*}^\top \end{bmatrix} \mathbf{e}_0' + \mathbf{p}_{\mathbf{e}}\right)}_{:=\mathbf{e}} \tag{3}$$

where $(\mathbf{B}^\top, \mathbf{B}^\top\mathbf{v} + \mathbf{e}_0')$ are some LWE samples and $\mathbf{p}_{\mathbf{e}} \in \mathbb{Z}^{2m}$ is a perturbation to ensure that no information about the trapdoor matrix $\mathbf{R}_{\mathsf{id}^*}$ is revealed. The re-randomization technique [KY16] guarantees that the simulated $\mathbf{e}$ is distributed statistically close a spherical Gaussian $D_{\mathbb{Z}^{2m},\sigma}$ under the condition that $\sigma$ is *linear* in $\|\mathbf{R}_{\mathsf{id}^*}\|$.

During the decryption step, the user who owns the secret key $\mathsf{sk}_{\mathsf{id}}$ can compute

$$c_0 - \mathbf{c}^\top \cdot \mathsf{sk}_{\mathsf{id}} = \left\lceil \frac{q}{2} \right\rceil \cdot \mu + \underbrace{e_0 - \langle \mathbf{e}, \mathsf{sk}_{\mathsf{id}} \rangle}_{:=\text{error term}}. \tag{4}$$

Since the Gaussian widths of $\mathsf{sk}_{\mathsf{id}}$ and $\mathbf{e}$ are both *linear* in $\|\mathbf{R}_{\mathsf{id}}\|$, the size of the error term is *quadratic* in $\|\mathbf{R}_{\mathsf{id}}\|$. To ensure decryption correctness, the modulus must exceed the size of the error term. Thus, the modulus is *quadratic* in $\|\mathbf{R}_{\mathsf{id}}\|$.

**Our new lattice IBE framework.** To remove this quadratic restriction, we propose a cross-multiplication structure. At a high level, our idea can be represented in Fig. 1.

[5] There exists an algorithm that, with input $\{\mathbf{B}\mathbf{R}_i + \kappa_i\mathbf{G}\}_{i\in[t]}$ and $\mathsf{id}$, homomorphically computes the partitioning function $H$ and outputs $\mathbf{B}\mathbf{R}_{\mathsf{id}} + H(\boldsymbol{\kappa}, \mathsf{id})\mathbf{G}$.

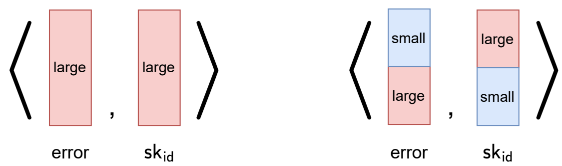


**Fig. 1.** The error term of previous works (left) and ours (right), where $\langle \cdot, \cdot \rangle$ represents vector inner product.

Let's look at Eqs. (2) and (3), and focus on the secret key $\mathsf{sk}_{\mathsf{id}}$ and the error $\mathbf{e}$—the two components that constitute the inner product in the error term. We split each of them into two parts: an upper half and a lower half.

$$\mathsf{sk}_{\mathsf{id}} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} := \mathbf{p}_{\mathsf{sk}} + \begin{bmatrix} -\mathbf{R}_{\mathsf{id}} \\ \mathbf{I} \end{bmatrix} \mathbf{z}, \quad \mathbf{e} = \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \end{bmatrix} := \mathbf{p}_{\mathbf{e}} + \begin{bmatrix} \mathbf{I} \\ \mathbf{R}_{\mathsf{id}^*}^\top \end{bmatrix} \mathbf{e}_0'.$$

We observe that the inner product between the secret key and the error has a natural cross-multiplication structure of the trapdoor $\mathbf{R}_{\mathsf{id}}$:

$\mathbf{R}_{\mathsf{id}}$ *only appears in the upper half* $\mathbf{x}_1$ *and the lower half* $\mathbf{e}_2$.

This leads to a key idea that we want to set $\mathbf{x}_1$ (upper) and $\mathbf{e}_2$ (lower) to be large, while keeping $\mathbf{x}_2$ (lower) and $\mathbf{e}_1$ (upper) small. Here, "large" means that the Gaussian width is linear in $\|\mathbf{R}_{\mathsf{id}}\|$. This idea allows us to fully utilize the natural cross-multiplication structure of the trapdoor $\mathbf{R}_{\mathsf{id}}$, and thus remove the quadratic restriction of modulus.

In the following, we describe how to obtain a $(D_{\mathbb{Z}^m,\theta_1}, D_{\mathbb{Z}^m,\theta_2})$-hybrid secret key $\left[\begin{smallmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{smallmatrix}\right]$ where $\theta_1 \gg \theta_2$, and a $(D_{\mathbb{Z}^m,\sigma_1}, D_{\mathbb{Z}^m,\sigma_2})$-hybrid error $\left[\begin{smallmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \end{smallmatrix}\right]$ where $\sigma_1 \ll \sigma_2$. Similar ideas have been used to improve signature size in [AGJ+24, Lemma 3.2] and [JHT22, Lemma 7], while we aim to reduce IBE modulus.

*Obtain a* $(D_{\mathbb{Z}^m,\theta_1}, D_{\mathbb{Z}^m,\theta_2})$*-hybrid secret key.* In the previous Gaussian convolution approach (e.g., [YJW23]), to ensure the secret key $\mathsf{sk}_{\mathsf{id}} := \mathbf{p}_{\mathsf{sk}} + \left[\begin{smallmatrix} -\mathbf{R}_{\mathsf{id}} \\ \mathbf{I} \end{smallmatrix}\right]\mathbf{z}$ is distributed statistically close to a *spherical* Gaussian distribution $D_{\mathbb{Z}^{2m},\theta}$, the perturbation $\mathbf{p}_{\mathsf{sk}}$ is sampled from a discrete Gaussian distribution with covariance matrix

$$\Sigma_p := \theta^2 \mathbf{I}_{2m} - \zeta^2 \begin{bmatrix} -\mathbf{R}_{\mathsf{id}} \\ \mathbf{I} \end{bmatrix} \begin{bmatrix} -\mathbf{R}_{\mathsf{id}}^\top \ \mathbf{I} \end{bmatrix},$$

where $\zeta$ denotes the Gaussian width of $\mathbf{z}$, and the parameters are set to satisfy $\theta = \tilde{O}\left(\zeta \cdot \|\left[\begin{smallmatrix} \mathbf{R}_{\mathsf{id}} \\ \mathbf{I} \end{smallmatrix}\right]\|\right)$. This requirement for $\theta$ comes from the following process. First, a middle term $\left[\begin{smallmatrix} -\mathbf{R}_{\mathsf{id}}^\top & \mathbf{I} \end{smallmatrix}\right](\theta^2 \mathbf{I}_{2m})\left[\begin{smallmatrix} -\mathbf{R}_{\mathsf{id}} \\ \mathbf{I} \end{smallmatrix}\right] = \theta^2 \cdot \left[\begin{smallmatrix} -\mathbf{R}_{\mathsf{id}}^\top & \mathbf{I} \end{smallmatrix}\right]\left[\begin{smallmatrix} -\mathbf{R}_{\mathsf{id}} \\ \mathbf{I} \end{smallmatrix}\right]$ is considered in [YJW23, Lemma 7]. Second, the above equation allows us to analyze the eigenvalue decomposition of $\left[\begin{smallmatrix} -\mathbf{R}_{\mathsf{id}}^\top & \mathbf{I} \end{smallmatrix}\right]\left[\begin{smallmatrix} -\mathbf{R}_{\mathsf{id}} \\ \mathbf{I} \end{smallmatrix}\right]$ separately. Third, the multiplication term $\theta^2\left[\begin{smallmatrix} -\mathbf{R}_{\mathsf{id}}^\top & \mathbf{I} \end{smallmatrix}\right]\left[\begin{smallmatrix} -\mathbf{R}_{\mathsf{id}} \\ \mathbf{I} \end{smallmatrix}\right]$ couples the parameter $\theta$ to the trapdoor matrix $\left[\begin{smallmatrix} -\mathbf{R}_{\mathsf{id}} \\ \mathbf{I} \end{smallmatrix}\right]$, thus we require $\theta^2$ to be larger than the largest eigenvalue of $\left[\begin{smallmatrix}-\mathbf{R}_{\mathsf{id}}^\top & \mathbf{I}\end{smallmatrix}\right]\left[\begin{smallmatrix}-\mathbf{R}_{\mathsf{id}} \\ \mathbf{I}\end{smallmatrix}\right]$ along with another factor $\zeta^2$, i.e., $\theta = \tilde{O}\left(\zeta \cdot \|\left[\begin{smallmatrix}\mathbf{R}_{\mathsf{id}} \\ \mathbf{I}\end{smallmatrix}\right]\|\right)$.

In our scheme, however, we instead require $\mathsf{sk}_{\mathsf{id}}$ is distributed statistically close to a *non-spherical* Gaussian distribution $D_{\mathbb{Z}^{2m},\sqrt{\Sigma}}$ where $\Sigma \coloneqq \left[\begin{smallmatrix}\theta_1^2\mathbf{I} & \\ & \theta_2^2\mathbf{I}\end{smallmatrix}\right]$. The key difference from the previous works is we require that only $\theta_1$ is linear in $\|\mathbf{R}_{\mathsf{id}}\|$ but $\theta_2$ is not. Accordingly, we sample the perturbation $\mathbf{p}_{\mathsf{sk}}$ from a discrete Gaussian distribution with covariance matrix

$$\Sigma_p \coloneqq \begin{bmatrix}\theta_1^2\mathbf{I} & \\ & \theta_2^2\mathbf{I}\end{bmatrix} - \zeta^2 \begin{bmatrix}-\mathbf{R}_{\mathsf{id}} \\ \mathbf{I}\end{bmatrix}\begin{bmatrix}-\mathbf{R}_{\mathsf{id}}^\top & \mathbf{I}\end{bmatrix} = \begin{bmatrix}\theta_1^2\mathbf{I} - \zeta^2\mathbf{R}_{\mathsf{id}}\mathbf{R}_{\mathsf{id}}^\top & \zeta^2\mathbf{R}_{\mathsf{id}} \\ \zeta^2\mathbf{R}_{\mathsf{id}}^\top & (\theta_2^2 - \zeta^2)\mathbf{I}\end{bmatrix}.$$

It seems that this new covariance matrix naturally leads to our desired parameter constraints, i.e., only $\theta_1$ is linear in $\|\mathbf{R}_{\mathsf{id}}\|$ while $\theta_2$ is just related to $\zeta$.

However, it is difficult to provide a direct proof for this covariance matrix. More precisely, the middle term becomes $\left[\begin{smallmatrix}-\mathbf{R}_{\mathsf{id}}^\top & \mathbf{I}\end{smallmatrix}\right]\left[\begin{smallmatrix}\theta_1^2\mathbf{I} & \\ & \theta_2^2\mathbf{I}\end{smallmatrix}\right]\left[\begin{smallmatrix}-\mathbf{R}_{\mathsf{id}} \\ \mathbf{I}\end{smallmatrix}\right]$ in our setting. The matrix multiplication is not commutative so we can no longer analyze the eigenvalue decomposition of $\left[\begin{smallmatrix}-\mathbf{R}_{\mathsf{id}}^\top & \mathbf{I}\end{smallmatrix}\right]\left[\begin{smallmatrix}-\mathbf{R}_{\mathsf{id}} \\ \mathbf{I}\end{smallmatrix}\right]$ separately. To solve this problem, we treat $\mathbf{R}_{\mathsf{id}}$ and $\mathbf{I}$ as two separate components. That is, we calculate our middle term in one more step, i.e.,

$$\begin{bmatrix}-\mathbf{R}_{\mathsf{id}}^\top & \mathbf{I}\end{bmatrix}\begin{bmatrix}\theta_1^2\mathbf{I} & \\ & \theta_2^2\mathbf{I}\end{bmatrix}\begin{bmatrix}-\mathbf{R}_{\mathsf{id}} \\ \mathbf{I}\end{bmatrix} = \theta_1^2 \cdot \mathbf{R}_{\mathsf{id}}^\top\mathbf{R}_{\mathsf{id}} + \theta_2^2\mathbf{I}.$$

Now $\mathbf{R}_{\mathsf{id}}^\top\mathbf{R}_{\mathsf{id}}$ appears as an individual term and then we can compute its eigenvalue decomposition separately. This method circumvents the need to compute the eigenvalue decomposition of $\left[\begin{smallmatrix}-\mathbf{R}_{\mathsf{id}}^\top & \mathbf{I}\end{smallmatrix}\right]\left[\begin{smallmatrix}-\mathbf{R}_{\mathsf{id}} \\ \mathbf{I}\end{smallmatrix}\right]$. Even better, this unique perspective allows us to decouple the parameter $\theta_2$ from the trapdoor $\mathbf{R}_{\mathsf{id}}$, leading to different ranges for parameters $\theta_1$ and $\theta_2$, i.e., $\theta_1 = \tilde{O}(\zeta \cdot \|\mathbf{R}_{\mathsf{id}}\|)$ and $\theta_2 = \tilde{O}(\zeta)$. Note that, $\theta_1$ achieves the same complexity as the previous $\theta$, and we do not introduce any extra terms into $\theta_2$.

*Obtain a $(D_{\mathbb{Z}^m,\sigma_1}, D_{\mathbb{Z}^m,\sigma_2})$-hybrid error.* The form of the error $\mathbf{e} \coloneqq \mathbf{p}_{\mathbf{e}} + \left[\begin{smallmatrix}\mathbf{I} \\ \mathbf{R}_{\mathsf{id}^*}^\top\end{smallmatrix}\right]\mathbf{e}_0'$ is essentially the same as the secret key. More precisely, to get an error $\mathbf{e}$ whose distribution is statistically close to a *non-spherical* Gaussian distribution $D_{\mathbb{Z}^{2m},\sqrt{\Sigma}}$ where $\Sigma \coloneqq \left[\begin{smallmatrix}\sigma_1^2\mathbf{I} & \\ & \sigma_2^2\mathbf{I}\end{smallmatrix}\right]$, we first sample the perturbation $\mathbf{p}_{\mathbf{e}}$ from a discrete Gaussian distribution with covariance matrix

$$\Sigma_p \coloneqq \begin{bmatrix}\sigma_1^2\mathbf{I} & \\ & \sigma_2^2\mathbf{I}\end{bmatrix} - \delta^2\begin{bmatrix}\mathbf{I} \\ \mathbf{R}_{\mathsf{id}^*}^\top\end{bmatrix}\begin{bmatrix}\mathbf{I} & \mathbf{R}_{\mathsf{id}^*}\end{bmatrix} = \begin{bmatrix}(\sigma_1^2 - \delta^2)\mathbf{I} & \delta^2\mathbf{R}_{\mathsf{id}^*} \\ \delta^2\mathbf{R}_{\mathsf{id}^*}^\top & \sigma_2^2 - \delta^2\mathbf{R}_{\mathsf{id}^*}^\top\mathbf{R}_{\mathsf{id}^*}\end{bmatrix},$$

where $\delta$ is the Gaussian width of the error $\mathbf{e}_0'$. In the derivation process, we use a similar trick, i.e., treating $\mathbf{R}_{\mathsf{id}^*}$ and $\mathbf{I}$ as two separate components instead of considering $\left[\begin{smallmatrix}\mathbf{I} \\ \mathbf{R}_{\mathsf{id}^*}^\top\end{smallmatrix}\right]$ as a single block, which yields the middle term

$$\begin{bmatrix}\mathbf{I} & \mathbf{R}_{\mathsf{id}^*}\end{bmatrix}\begin{bmatrix}\sigma_1^2\mathbf{I} & \\ & \sigma_2^2\mathbf{I}\end{bmatrix}\begin{bmatrix}\mathbf{I} \\ \mathbf{R}_{\mathsf{id}^*}^\top\end{bmatrix} = \sigma_1^2\mathbf{I} + \sigma_2^2 \cdot \mathbf{R}_{\mathsf{id}^*}\mathbf{R}_{\mathsf{id}^*}^\top.$$

This equation allows us to bypass the problem of not being able to analyze the eigenvalue decomposition of $[\mathbf{I}\ \mathbf{R}_{\mathsf{id}^*}]\begin{bmatrix}\mathbf{I}\\ \mathbf{R}_{\mathsf{id}^*}^\top\end{bmatrix}$ separately, and simultaneously decouple the parameter $\sigma_1$ from the trapdoor $\mathbf{R}_{\mathsf{id}^*}$ and achieve our desired parameter constraints, i.e., $\sigma_1 = \tilde{O}(\delta)$ and $\sigma_2 = \tilde{O}(\delta \cdot \|\mathbf{R}_{\mathsf{id}^*}\|)$. Note that, $\sigma_2$ achieves the same complexity as the previous $\sigma$ (which is required $\sigma = \tilde{O}(\delta \cdot \|\begin{bmatrix}\mathbf{I}\\ \mathbf{R}_{\mathsf{id}^*}^\top\end{bmatrix}\|)$ by the re-randomization algorithm in [KY16]), and we do not introduce any extra terms into $\sigma_1$.

Therefore, we obtain a $(D_{\mathbb{Z}^m,\theta_1}, D_{\mathbb{Z}^m,\theta_2})$-hybrid secret key where $\theta_1 \gg \theta_2$ and a $(D_{\mathbb{Z}^m,\sigma_1}, D_{\mathbb{Z}^m,\sigma_2})$-hybrid error where $\sigma_1 \ll \sigma_2$, only the big ones $\theta_1$ and $\sigma_2$ are *linear* in $\|\mathbf{R}_{\mathsf{id}}\|$, but the small ones $\theta_2$ and $\sigma_1$ are not. Consequently, during the decryption step, the user who owns the secret key $\mathsf{sk}_{\mathsf{id}}$ can compute

$$c_0 - \mathbf{c}^\top \cdot \mathsf{sk}_{\mathsf{id}} = \left\lceil \frac{q}{2} \right\rceil \cdot \mu + \underbrace{e_0 - \left\langle \begin{bmatrix}\mathbf{e}_1\\ \mathbf{e}_2\end{bmatrix}, \begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix} \right\rangle}_{:=\text{error term}},$$

where $\begin{bmatrix}\mathbf{e}_1\\ \mathbf{e}_2\end{bmatrix}$ is a $(D_{\mathbb{Z}^m,\sigma_1}, D_{\mathbb{Z}^m,\sigma_2})$-hybrid error and $\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}$ is a $(D_{\mathbb{Z}^m,\theta_1}, D_{\mathbb{Z}^m,\theta_2})$-hybrid secret key. Compared to Eq. (4), our new error term achieves a cross-inner-product structure as shown in Fig. 1. Thus, we remove the quadratic restriction.

## 3 Preliminaries

*Notations.* We denote $\mathbb{Z}, \mathbb{N}$ and $\mathbb{R}$ as the set of integers, the set of natural numbers and the set of real numbers, respectively. We use bold uppercase letters (e.g., $\mathbf{A}$) to denote matrices, and bold lowercase letters (e.g., $\mathbf{a}$) for column vectors. We use $\|\mathbf{a}\|$ to denote the Euclidean norm of vector $\mathbf{a}$ and denote $\|\mathbf{A}\| := \sup_{\|\mathbf{x}\|=1} \|\mathbf{A}\mathbf{x}\|$ the spectral norm of $\mathbf{A}$. We denote the horizontal concatenation of two vectors $\mathbf{a}, \mathbf{b}$ by $[\mathbf{a}^\top|\mathbf{b}^\top]$. We use $\widetilde{\mathbf{A}}$ to denote the Gram-Schmidt orthogonalization of $\mathbf{A}$. We write $\Sigma \succ 0$, when a symmetric matrix $\Sigma \in \mathbb{R}^{m\times m}$ is positive definite, i.e., $\mathbf{x}^\top \Sigma \mathbf{x} > 0$ for all nonzero $\mathbf{x} \in \mathbb{R}^m$. For any scalar $s$, we write $\Sigma \succ s$ if $\Sigma - s \cdot \mathbf{I} \succ 0$.

*Lattices.* An $n$-dimensional (full rank) lattice $\Lambda \subseteq \mathbb{R}^n$ is the set of all integer linear combinations of some set of $n$ linearly independent basis vectors $\mathbf{B} = \{\mathbf{b}_1, \cdots, \mathbf{b}_n\} \subseteq \mathbb{R}^n$, $\Lambda = \{\mathbf{B}\mathbf{x} \mid \mathbf{x} \in \mathbb{Z}^n\}$. We denote the dual lattice of $\Lambda$ by $\Lambda^* := \{\mathbf{y} \in \mathrm{span}(\Lambda) \mid \langle \mathbf{y}, \mathbf{x}\rangle \in \mathbb{Z}, \forall \mathbf{x} \in \Lambda\}$. For a matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$, define $\Lambda_q^\perp(\mathbf{A}) := \{\mathbf{e} \in \mathbb{Z}^m : \mathbf{A}\mathbf{e} = \mathbf{0} \bmod q\}$, which is a full-rank $m$-dimensional integer lattice. We omit $q$ when it is clear from the context.

*Gaussian distributions.* The Gaussian function $\rho : \mathbb{R}^m \to (0, 1]$ is defined as $\rho(\mathbf{x}) = \exp(-\pi \cdot \langle \mathbf{x}, \mathbf{x}\rangle)$. Applying a linear transformation given by an invertible matrix $\mathbf{B}$ yields $\rho_{\mathbf{B}}(\mathbf{x}) = \rho(\mathbf{B}^{-1}\mathbf{x}) = \exp(-\pi \cdot \mathbf{x}^\top \Sigma^{-1}\mathbf{x})$, where $\Sigma = \mathbf{B}\mathbf{B}^\top$. Since $\rho_{\mathbf{B}}$ is exactly determined by $\Sigma$, we also write it as $\rho_{\sqrt{\Sigma}}$. We denote $D_{\sqrt{\Sigma}}$ as the continuous Gaussian distribution over $\mathbb{R}^m$ where the probability density function $D_{\sqrt{\Sigma}}(\mathbf{x}) \propto \rho_{\mathbf{B}}(\mathbf{x})$. For a lattice $\Lambda$ and $\mathbf{c} \in \operatorname{span}(\Lambda)$, the discrete Gaussian distribution $D_{\Lambda+\mathbf{c},\sqrt{\Sigma}}$ is defined as: for any $\mathbf{x} \in \Lambda + \mathbf{c}$,

$$D_{\Lambda+\mathbf{c},\sqrt{\Sigma}}(\mathbf{x}) = \frac{\rho_{\sqrt{\Sigma}}(\mathbf{x})}{\rho_{\sqrt{\Sigma}}(\Lambda + \mathbf{c})}.$$

When $\Sigma = \sigma^2\mathbf{I}$ for some $\sigma > 0$, we write $\rho_{\sqrt{\Sigma}}$, $D_{\sqrt{\Sigma}}$, $D_{\Lambda+\mathbf{c},\sqrt{\Sigma}}$ as $\rho_\sigma$, $D_\sigma$, $D_{\Lambda+\mathbf{c},\sigma}$.

We recall the definition of the smoothing parameter of lattices as follows.

**Definition 1 ([MR07]).** *For any $\epsilon > 0$, any $n$-dimensional lattice $\Lambda$, the smoothing parameter $\eta_\epsilon(\Lambda)$ is the smallest real $s > 0$ such that $\rho_{1/s}(\Lambda^*) \leq 1 + \epsilon$. We write $\sqrt{\Sigma} \geq \eta_\epsilon(\Lambda)$ if $\rho_{\sqrt{\Sigma^{-1}}}(\Lambda^*) \leq 1 + \epsilon$.*

For the discrete Gaussian over lattices, we have the following tail bounds.

**Lemma 1 ([MP12]).** *For $\sigma > \eta_\epsilon(\mathbb{Z})$, $t \geq 0$, $\mathsf{Pr}_{x \leftarrow D_{\mathbb{Z},\sigma}}[|x| \geq t] \leq 2e^{-\pi \cdot \frac{t^2}{\sigma^2}}$.*

**Lemma 2 ([KY16]).** *For $\sigma > \eta_\epsilon(\mathbb{Z}^n)$, $t \geq 0$, $\mathbf{x} \in \mathbb{Z}^n$, $\Pr_{\mathbf{e} \leftarrow D_{\mathbb{Z}^n,\sigma}}[|\mathbf{e}^\top \mathbf{x}| \geq t] \leq 2e^{-\pi \cdot \frac{t^2}{\|\mathbf{x}\|^2 \sigma^2}}$.*

**Lemma 3 ([MR07]).** *For $n$-dimensional lattice $\Lambda$, $\epsilon \in (0, \frac{1}{2})$, $s \geq \eta_\epsilon(\Lambda)$, we have $\Pr_{\mathbf{x} \leftarrow D_{\Lambda,s}}[\|\mathbf{x}\| > s\sqrt{n}] \leq 2^{-n}$.*

**Lemma 4 ([GMPW20]).** *For any $\epsilon \in [0, 1)$, defining $\bar{\epsilon} = \frac{2\epsilon}{1-\epsilon}$, matrix $\mathbf{S}$ of full column rank, lattice coset $A = \Lambda + \mathbf{a} \subset \operatorname{span}(\mathbf{S})$, and matrix $\mathbf{T}$ such that $\ker(\mathbf{T})$ is a $\Lambda$-subspace and $\eta_\epsilon(\Lambda \cap \ker(\mathbf{T})) \leq \mathbf{S}$, the statistical distance between $\mathbf{T} \cdot D_{A,\mathbf{S}}$ and $D_{\mathbf{T}A,\mathbf{TS}}$ is at most $\bar{\epsilon}$.*

*Trapdoor generation and Gaussian sampling.* Lattices admit natural and innate "trapdoors" that have many useful cryptographic applications. As observed in [Ajt96,MP12,DLP14], a short basis of a lattice could serve as such a trapdoor. In another line, Gentry et al. [GPV08] showed how to use a short basis to sample lattice points from a discrete Gaussian probability distribution.

**Lemma 5 ([Ajt96, AP09]).** *Let $m \geq 6n\lceil \log q \rceil$. There exists an efficient trapdoor generation algorithm $\mathsf{TrapGen}(1^n, 1^m, q)$ that outputs a full-rank matrix $\mathbf{B} \in \mathbb{Z}_q^{n \times m}$ and a basis $\mathbf{T_B} \in \mathbb{Z}^{m \times m}$ for $\Lambda_q^\perp(\mathbf{B})$ such that $\mathbf{B}$ is statistically close to uniform and $\|\widetilde{\mathbf{T_B}}\| = O(\sqrt{n \log q})$ with overwhelming probability.*

**Lemma 6 ([MP12]).** *For positive integers $b, q$ and $k > k' \geq \lceil \log_b q \rceil$, let $\mathbf{g}_b^\top = [1|b|b^2|\cdots|b^{k'}|\mathbf{0}] \in \mathbb{Z}^k$ be the gadget vector, then the gadget matrix $\mathbf{G}_b$ is defined as $\mathbf{G}_b \coloneqq \mathbf{I}_n \otimes \mathbf{g}_b^\top \in \mathbb{Z}^{n \times m}$. We ignore the subscript if $b = 2$. The lattice $\Lambda_q^\perp(\mathbf{G}_b)$ has a publicly known basis $\mathbf{T}_{\mathbf{G}_b} \in \mathbb{Z}^{m \times m}$ with $\|\widetilde{\mathbf{T}_{\mathbf{G}_b}}\| \leq \sqrt{b^2 + 1}$.*

**Lemma 7 ([GPV08]).** *For a trapdoor pair $(\mathbf{B}, \mathbf{T_B}) \in \mathbb{Z}_q^{n \times m} \times \mathbb{Z}^{m \times m}$ such that $\mathbf{B} \cdot \mathbf{T_B} = \mathbf{0} \bmod q$ and $\mathbf{B}$ is a full-rank matrix, a positive real $\sigma \geq \eta_\epsilon(\mathbb{Z}^m) \cdot \|\widetilde{\mathbf{T_B}}\|$ and a given target $\mathbf{u} \in \mathbb{Z}_q^n$, there exists a preimage sampling algorithm $\mathsf{SamplePre}(\mathbf{B}, \mathbf{T_B}, \mathbf{u}, \sigma)$ that outputs a vector $\mathbf{x}$ such that $\mathbf{Bx} = \mathbf{u} \bmod q$ and the distribution of $\mathbf{x}$ is statistically close to $D_{\mathbb{Z}^m,\sigma}$ conditioned on $\mathbf{Bx} = \mathbf{u} \bmod q$.*

*Leftover Hash Lemma (LHL).* As observed in [Reg05, ABB10], the following lemma is obtained as a corollary to the (general) leftover hash lemma.

**Lemma 8.** *Let $q \in \mathbb{N}$ be an odd prime and let $m > (n+1)\log q + \omega(\log n)$. Let $\mathbf{R} \xleftarrow{\$} \{-1,1\}^{m\times m}$ and $\mathbf{B}, \mathbf{B}' \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$ be uniformly random matrices. Then the distribution of $(\mathbf{B}, \mathbf{BR})$ is statistically close to the distribution of $(\mathbf{B}, \mathbf{B}')$.*

*Learning With Errors (LWE) Assumption.* We recall the LWE problem, which was introduced by Regev [Reg05].

**Definition 2 (LWE).** *For an integer $n = n(\lambda)$, $m = m(n)$, a prime integer $q = q(n) > 2$, and a PPT algorithm $\mathcal{A}$, the advantage of $\mathcal{A}$ for the learning with errors problem $\mathsf{LWE}_{n,m,q,\delta}$ is defined as follows:*

$$\mathsf{Adv}_{\mathcal{A}}^{\mathsf{LWE}_{n,m,q,\delta}}(\lambda) := |\Pr[\mathcal{A}(\mathbf{B}, \mathbf{v}^\top\mathbf{B} + \mathbf{e}^\top) \to 1] - \Pr[\mathcal{A}(\mathbf{B}, \mathbf{y}^\top + \mathbf{e}^\top) \to 1]|$$

*where $\mathbf{B} \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$, $\mathbf{v} \xleftarrow{\$} \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m,\delta}$, $\mathbf{y} \xleftarrow{\$} \mathbb{Z}_q^m$. We say that $\mathsf{LWE}_{n,m,q,\delta}$ assumption holds if $\mathsf{Adv}_{\mathcal{A}}^{\mathsf{LWE}_{n,m,q,\delta}}(\lambda)$ is negligible for all PPT $\mathcal{A}$.*

Regev [Reg05] showed that solving $\mathsf{LWE}_{n,m,q,\delta}$ for $\delta > 2\sqrt{2n}$ is (quantumly) as hard as approximating the SIVP and GapSVP problems to within $\tilde{O}(nq/\delta)$ factors in the $\ell_2$ norm, in the worst case. Later, (partial) dequantumization of the Regev's reduction was achieved [Pei09].

### 3.1 Identity-Based Encryption

*Syntax.* We recall the standard syntax of IBE [Sha84, BF01] in the following. An identity-based encryption scheme $\Pi$ with identity space $\mathcal{ID}$ consists of four PPT algorithms $(\mathsf{Setup}, \mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ as follows.

- $\mathsf{Setup}(1^\lambda)$: Given the security parameter $\lambda$, it outputs the master public key $\mathsf{mpk}$ and the master secret key $\mathsf{msk}$.
- $\mathsf{KeyGen}(\mathsf{mpk}, \mathsf{msk}, \mathsf{id})$: Given $(\mathsf{mpk}, \mathsf{msk})$ and an identity $\mathsf{id} \in \mathcal{ID}$, it outputs the secret key $\mathsf{sk}_{\mathsf{id}}$.
- $\mathsf{Enc}(\mathsf{mpk}, \mathsf{id}, \mu)$: Given the master public key $\mathsf{mpk}$, an identity $\mathsf{id} \in \mathcal{ID}$, and a message $\mu$, it outputs a ciphertext $\mathsf{ct}$.
- $\mathsf{Dec}(\mathsf{mpk}, \mathsf{sk}_{\mathsf{id}}, \mathsf{ct})$: Given the master public key $\mathsf{mpk}$, the secret key $\mathsf{sk}_{\mathsf{id}}$, and a ciphertext $\mathsf{ct}$, it outputs a message $\mu'$ or $\perp$.

*Correctness.* We say an IBE scheme $\Pi$ is correct, if for all $\mathsf{id} \in \mathcal{ID}$ and all $\mu$ in the specified message space, it holds that $\Pr[\mathsf{Dec}(\mathsf{mpk}, \mathsf{sk}_{\mathsf{id}}, \mathsf{ct}) \neq \mu] = \mathsf{negl}(\lambda)$, where the probability is taken over the randomness used in $(\mathsf{mpk}, \mathsf{msk}) \xleftarrow{\$} \mathsf{Setup}(1^\lambda)$, $\mathsf{sk}_{\mathsf{id}} \xleftarrow{\$} \mathsf{KeyGen}(\mathsf{mpk}, \mathsf{msk}, \mathsf{id})$ and $\mathsf{ct} \xleftarrow{\$} \mathsf{Enc}(\mathsf{mpk}, \mathsf{id}, \mu)$.

*Adaptive anonymous security.* We consider the adaptive anonymous security notion for IBE as in [Yam17]. This security is defined by the following game between a challenger and an adversary $\mathcal{A}$.

Setup: At the beginning of the game, the challenger runs $\mathsf{Setup}(1^\lambda)$ to get $(\mathsf{mpk}, \mathsf{msk})$ and sends $\mathsf{mpk}$ to $\mathcal{A}$.

Phase 1: $\mathcal{A}$ may adaptively make key extraction queries. When $\mathcal{A}$ submits $\mathsf{id}$, the challenger returns $\mathsf{sk}_{\mathsf{id}} \stackrel{\$}{\leftarrow} \mathsf{KeyGen}(\mathsf{mpk}, \mathsf{msk}, \mathsf{id})$.

Challenge: At some point, $\mathcal{A}$ outputs a message $\mu$ and an identity $\mathsf{id}^*$, on which it wishes to be challenged. Then, the challenger picks a random bit $\mathsf{coin} \stackrel{\$}{\leftarrow} \{0,1\}$ and a random ciphertext $\mathsf{ct}_1^* \leftarrow \mathcal{C}$ from the ciphertext space. If $\mathsf{coin} = 0$, it runs $\mathsf{ct}_0^* \stackrel{\$}{\leftarrow} \mathsf{Enc}(\mathsf{mpk}, \mathsf{id}^*, \mu)$ and gives the challenge ciphertext $\mathsf{ct}_0^*$ to $\mathcal{A}$. If $\mathsf{coin} = 1$, it gives $\mathsf{ct}_1^*$ to $\mathcal{A}$.

Phase 2: $\mathcal{A}$ continues to make key queries with a restriction that $\mathsf{id} \neq \mathsf{id}^*$.

Guess: Finally, $\mathcal{A}$ outputs a bit $\widehat{\mathsf{coin}}$.

The advantage of $\mathcal{A}$ is defined as $|\Pr[\widehat{\mathsf{coin}} = \mathsf{coin}] - \frac{1}{2}|$. We say that the scheme satisfies adaptively-anonymous security if the advantage of any PPT $\mathcal{A}$ is negligible.

*Partitioning functions.* In the following, we recall the notion of the partition function. This notion abstracts out the information-theoretic properties that are useful in the security proofs based on the partitioning techniques.

**Definition 3 ([Yam17]).** *Let $H := \{H_\lambda : \mathcal{K}_\lambda \times \mathcal{ID}_\lambda \to \mathcal{T}_\lambda\}$ be an ensemble of function families. We say that $H$ is a partitioning function, if there exists an efficient algorithm $\mathsf{PrtSmp}(1^\lambda, Q, \epsilon)$, which takes as input polynomially bounded $Q = Q(\lambda) \in \mathbb{N}$ and noticeable $\epsilon = \epsilon(\lambda) \in (0, \frac{1}{2}]$ outputs $\boldsymbol{\kappa}$ such that*

1. *There exists $\lambda_0 \in \mathbb{N}$ such that for any $\lambda > \lambda_0$*
$$\Pr\left[\boldsymbol{\kappa} \in \mathcal{K}_\lambda : \boldsymbol{\kappa} \stackrel{\$}{\leftarrow} \mathsf{PrtSmp}\left(1^\lambda, Q(\lambda), \epsilon(\lambda)\right)\right] = 1.$$
*Here, $\lambda_0$ may depend on functions $Q(\lambda)$ and $\epsilon(\lambda)$.*
2. *For $\lambda > \lambda_0$, there exists $\gamma_{\max}(\lambda)$ and $\gamma_{\min}(\lambda)$ that depend on $Q(\lambda)$ and $\epsilon(\lambda)$ such that for all $\mathsf{id}^{(1)}, \cdots, \mathsf{id}^{(Q)}, \mathsf{id}^* \in \mathcal{ID}_\lambda$ with $\mathsf{id}^* \notin \{\mathsf{id}^{(1)}, \cdots, \mathsf{id}^{(Q)}\}$,*
$$\gamma_{\max}(\lambda) \geq \Pr[H(\boldsymbol{\kappa}, \mathsf{id}^{(1)}) \neq 0 \wedge \cdots \wedge H(\boldsymbol{\kappa}, \mathsf{id}^{(Q)}) \neq 0 \wedge H(\boldsymbol{\kappa}, \mathsf{id}^*) = 0] \geq \gamma_{\min}(\lambda)$$
*holds and the probability is taken over the choice of $\boldsymbol{\kappa} \stackrel{\$}{\leftarrow} \mathsf{PrtSmp}\big(1^\lambda, Q(\lambda), \epsilon(\lambda)\big)$. Moreover, the function $\tau(\lambda)$ defined as*
$$\tau(\lambda) := \gamma_{\min}(\lambda) \cdot \epsilon(\lambda) - \frac{\gamma_{\max}(\lambda) - \gamma_{\min}(\lambda)}{2}$$
*is noticeable.*

*We call $\boldsymbol{\kappa}$ the partitioning key and $\tau(\lambda)$ the quality of the partitioning function.*

The lattice IBE framework [ABB10, KY16, Yam17] requires to design two *deterministic* ways to homomorphically compute the partitioning function, which can be formally captured by the notion of $\Delta$-expanding algorithm.

**Definition 4 ([Yam17]).** *The deterministic algorithms* $(\mathsf{PubEval}, \mathsf{TrapEval})$ *are $\Delta$-expanding with respect to a partitioning function family $\big\{H : \mathcal{K} \times \mathcal{ID} \to \mathcal{T}\big\}$, if they are efficient and satisfy the following properties:*

- $\mathsf{PubEval}\big(\{\mathbf{C}_i \in \mathbb{Z}_q^{n\times m}\}_{i\in[t]}, \mathsf{id} \in \mathcal{ID}\big) \to \mathbf{C}_{\mathsf{id}} \in \mathbb{Z}_q^{n\times m}$.
- $\mathsf{TrapEval}\big(\mathbf{B} \in \mathbb{Z}_q^{n\times m}, \{\mathbf{R}_i \in \mathbb{Z}^{m\times m}\}_{i\in[t]}, \boldsymbol{\kappa} \in \mathcal{K}, \mathsf{id} \in \mathcal{ID}\big) \to \mathbf{R}_{\mathsf{id}} \in \mathbb{Z}^{m\times m}$.

*We require that the following holds:*

$$\mathsf{PubEval}(\{\mathbf{B}\mathbf{R}_i + \kappa_i\mathbf{G}\}_{i\in[t]}, \mathsf{id}) = \mathbf{B}\mathbf{R}_{\mathsf{id}} + H(\boldsymbol{\kappa}, \mathsf{id}) \cdot \mathbf{G}.$$

*where $\boldsymbol{\kappa} := (\kappa_1, \cdots, \kappa_t)$. Furthermore, we have $\|\mathbf{R}_{\mathsf{id}}\| \le \Delta \cdot \max_{i\in[t]} \|\mathbf{R}_i\|$.*

The following lemma addresses a general statement for bounding the success probability of an adversary engaging with the security game of IBE using the partitioning technique.

**Lemma 9 ([KY16]).** *Let us consider an IBE scheme and an adversary $\mathcal{A}$ that breaks the adaptively-anonymous security with advantage $\epsilon$. Let the identity space be $\mathcal{ID}$ and consider a map $\gamma(\cdot)$ that maps a sequence of identities in $\mathcal{ID}$ to a value in $[0, 1)$. We consider the following experiment. We first execute the security game for $\mathcal{A}$. Let $\mathsf{id}^*$ be the challenge identity and $\mathsf{id}^{(1)}, \cdots, \mathsf{id}^{(Q)}$ be the identities for which key extraction queries were made. We denote $\mathbb{ID} := (\mathsf{id}^*, \mathsf{id}^{(1)}, \cdots, \mathsf{id}^{(Q)})$. At the end of the game, we set $\mathsf{coin}' = \widehat{\mathsf{coin}}$ with probability $\gamma(\mathbb{ID})$ and $\mathsf{coin}' \xleftarrow{\$} \{0, 1\}$ with probability $1 - \gamma(\mathbb{ID})$. Then, the following holds*

$$\left|\Pr[\mathsf{coin}' = \mathsf{coin}] - \frac{1}{2}\right| \ge \gamma_{\min} \cdot \epsilon - \frac{\gamma_{\max} - \gamma_{\min}}{2},$$

*where $\gamma_{\min}$ (resp. $\gamma_{\max}$) is the maximum (resp. minimum) of $\gamma(\mathbb{ID})$ taken over all possible $\mathbb{ID}$.*

## 4 Sampling Algorithms with Non-Spherical Outputs

In this section, we follow the idea of the sampling algorithms in [ABB10] and design new sampling algorithms with non-spherical outputs.

### 4.1 Recall sampling algorithms in [ABB10] with spherical outputs

Agrawal et al. [ABB10] proposed two sampling algorithms to achieve efficient adaptively secure IBE in the standard model. More specifically, they constructed a family of lattices for which there are two distinct trapdoors for sampling, one is the Ajtai trapdoor pair, consisting of a matrix $\mathbf{B} \in \mathbb{Z}_q^{n\times m}$ and its associated trapdoor $\mathbf{T_B} \in \mathbb{Z}^{m\times m}$ satisfying $\mathbf{B} \times \mathbf{T_B} = \mathbf{0} \bmod q$, another is a small-norm matrix $\mathbf{R} \in \mathbb{Z}^{m\times m}$ with random entries. To accommodate these two trapdoors, they design two corresponding sampling algorithms $\mathsf{SampleLeft}$ and $\mathsf{SampleRight}$, respectively. Given a uniform random vector $\mathbf{u} \in \mathbb{Z}_q^n$, the outputs of both algorithms are statistically close to a discrete Gaussian distribution $D_{\mathbb{Z}^{2m},\theta}$.

$\mathsf{SampleLeft}(\mathbf{B}, \mathbf{M} \in \mathbb{Z}_q^{n\times m}, \mathbf{T_B} \in \mathbb{Z}^{m\times m}, \mathbf{u} \in \mathbb{Z}_q^n, \theta \in \mathbb{R})$

1. sample $\mathbf{x}_2 \leftarrow D_{\mathbb{Z}^m,\theta}$.
2. run $\mathbf{x}_1 \leftarrow \mathsf{SamplePre}(\mathbf{B}, \mathbf{T_B}, \mathbf{u}', \theta)$ where $\mathbf{u}' \coloneqq \mathbf{u} - \mathbf{M}\mathbf{x}_2$.
3. output $\mathbf{x} \coloneqq \left[\begin{smallmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{smallmatrix}\right]$.

$\mathsf{SampleRight}(\mathbf{B}, \mathbf{G} \in \mathbb{Z}_q^{n\times m}, \mathbf{T_G}, \mathbf{R} \in \mathbb{Z}^{m\times m}, H \in \mathbb{Z}_q^*, \mathbf{u} \in \mathbb{Z}_q^n, \theta, \zeta \in \mathbb{R})$

1. sample $\mathbf{p} \leftarrow D_{\mathbb{Z}^{2m},\sqrt{\Sigma_p}}$ where $\Sigma_p \coloneqq \theta^2\mathbf{I}_{2m} - \zeta^2\left[\begin{smallmatrix}-\mathbf{R}\\ \mathbf{I}_m\end{smallmatrix}\right]\left[\begin{smallmatrix}-\mathbf{R}^\top & \mathbf{I}_m\end{smallmatrix}\right]$.
2. run $\mathbf{z} \leftarrow \mathsf{SamplePre}(\mathbf{G}, \mathbf{T_G}, \mathbf{u}'', \zeta)$ where $\mathbf{u}'' \coloneqq H^{-1} \cdot (\mathbf{u} - [\mathbf{B}|\mathbf{BR} + H\mathbf{G}]\mathbf{p})$.
3. output $\mathbf{x} \coloneqq \mathbf{p} + \left[\begin{smallmatrix}-\mathbf{R}\\ \mathbf{I}\end{smallmatrix}\right]\mathbf{z}$.

The parameters $\zeta$ and $\theta$ should satisfy $\zeta \geq \eta_\epsilon(\mathbb{Z}^m) \cdot \|\widetilde{\mathbf{T_G}}\|$, $\theta \geq \eta_\epsilon(\mathbb{Z}^m) \cdot \|\widetilde{\mathbf{T_B}}\|$ and $\theta^2 \geq (\zeta^2 + \eta_\epsilon^2(\mathbb{Z}^m)) \cdot \|\mathbf{R}\|^2 + 2\zeta^2 + 2\eta_\epsilon^2(\mathbb{Z}^m)$.

### 4.2 Our new sampling algorithms with non-spherical outputs

To generate non-spherical outputs, i.e., a discrete Gaussian distribution $D_{\mathbb{Z}^{2m},\sqrt{\Sigma}}$ where $\Sigma \coloneqq \left[\begin{smallmatrix}\theta_1^2\mathbf{I} & \\ & \theta_2^2\mathbf{I}\end{smallmatrix}\right]$ where $\theta_1 \gg \theta_2$, we design new sampling algorithms $\mathsf{SampleLeft}_{\textsf{non-sp}}$ and $\mathsf{SampleRight}_{\textsf{non-sp}}$ as follows.

$\mathsf{SampleLeft}_{\textsf{non-sp}}(\mathbf{B}, \mathbf{M} \in \mathbb{Z}_q^{n\times m}, \mathbf{T_B} \in \mathbb{Z}^{m\times m}, \mathbf{u} \in \mathbb{Z}_q^n, \theta_1, \theta_2 \in \mathbb{R})$

1. sample $\mathbf{x}_2 \leftarrow D_{\mathbb{Z}^m,\theta_2}$.
2. run $\mathbf{x}_1 \leftarrow \mathsf{SamplePre}(\mathbf{B}, \mathbf{T_B}, \mathbf{u}', \theta_1)$ where $\mathbf{u}' \coloneqq \mathbf{u} - \mathbf{M}\mathbf{x}_2$.
3. output $\mathbf{x} \coloneqq \left[\begin{smallmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{smallmatrix}\right]$.

$\mathsf{SampleRight}_{\textsf{non-sp}}(\mathbf{B}, \mathbf{G} \in \mathbb{Z}_q^{n\times m}, \mathbf{T_G}, \mathbf{R} \in \mathbb{Z}^{m\times m}, H \in \mathbb{Z}_q^*, \mathbf{u} \in \mathbb{Z}_q^n, \theta_1, \theta_2, \zeta \in \mathbb{R})$

1. sample $\mathbf{p} \leftarrow D_{\mathbb{Z}^{2m},\sqrt{\Sigma_p}}$ where $\Sigma_p \coloneqq \left[\begin{smallmatrix}\theta_1^2\mathbf{I} & \\ & \theta_2^2\mathbf{I}\end{smallmatrix}\right] - \zeta^2\left[\begin{smallmatrix}-\mathbf{R}\\ \mathbf{I}\end{smallmatrix}\right]\left[\begin{smallmatrix}-\mathbf{R}^\top & \mathbf{I}\end{smallmatrix}\right]$.
2. run $\mathbf{z} \leftarrow \mathsf{SamplePre}(\mathbf{G}, \mathbf{T_G}, \mathbf{u}'', \zeta)$ where $\mathbf{u}'' \coloneqq H^{-1} \cdot (\mathbf{u} - [\mathbf{B}|\mathbf{BR} + H\mathbf{G}]\mathbf{p})$.
3. output $\mathbf{x} \coloneqq \mathbf{p} + \left[\begin{smallmatrix}-\mathbf{R}\\ \mathbf{I}\end{smallmatrix}\right]\mathbf{z}$.

In the following theorem, we prove that the outputs of our new sampling algorithms $\mathsf{SampleLeft}_{\textsf{non-sp}}$ and $\mathsf{SampleRight}_{\textsf{non-sp}}$ are statistically close to a non-spherical Gaussian.

**Theorem 1.** *Let $n, m$ be positive integers and $q$ be some prime. For a trapdoor pair $(\mathbf{B}, \mathbf{T_B}) \in \mathbb{Z}_q^{n\times m} \times \mathbb{Z}^{m\times m}$ (Lemma 5), a public gadget pair $(\mathbf{G}, \mathbf{T_G}) \in \mathbb{Z}_q^{n\times m} \times \mathbb{Z}^{m\times m}$ (Lemma 6), two matrices $\mathbf{M} \in \mathbb{Z}_q^{n\times m}$ and $\mathbf{R} \in \mathbb{Z}^{m\times m}$, a vector $\mathbf{u} \in \mathbb{Z}_q^n$, a non-zero (i.e., invertible) element $H \in \mathbb{Z}_q^*$, three positive reals $\zeta, \theta_1, \theta_2$ such that $\zeta \geq \eta_\epsilon(\mathbb{Z}^m) \cdot \|\widetilde{\mathbf{T_G}}\|$, $\theta_1^2 \geq (\zeta^2 + \eta_\epsilon^2(\mathbb{Z}^m))\|\mathbf{R}\|^2 + 2\zeta^2 + 4\eta_\epsilon^2(\mathbb{Z}^m)$, $\theta_1 \geq \sqrt{2}\zeta\|\mathbf{R}\|$ and $\theta_1 \geq \eta_\epsilon(\mathbb{Z}^m) \cdot \|\widetilde{\mathbf{T_B}}\|$, $\theta_2 > \sqrt{2}\zeta$, let $\Sigma := \begin{bmatrix} \theta_1^2\mathbf{I} & \\ & \theta_2^2\mathbf{I}\end{bmatrix}$, then the output of* $\mathsf{SampleLeft}_{\mathsf{non\text{-}sp}}$ *is statistically close to $D_{\mathbb{Z}^{2m},\sqrt{\Sigma}}$ conditioned on $[\mathbf{B}|\mathbf{M}]\mathbf{x} = \mathbf{u}$ and the output of* $\mathsf{SampleRight}_{\mathsf{non\text{-}sp}}$ *is statistically close to $D_{\mathbb{Z}^{2m},\sqrt{\Sigma}}$ conditioned on $[\mathbf{B}|\mathbf{BR} + H\mathbf{G}]\mathbf{x} = \mathbf{u}$.*

*Proof.* It is easy to check that by Lemma 7, the output of $\mathsf{SampleLeft}_{\mathsf{non\text{-}sp}}$ follows a non-spherical Gaussian whose covariance matrix is $\begin{bmatrix} \theta_1^2\mathbf{I} & \\ & \theta_2^2\mathbf{I}\end{bmatrix}$, when $\theta_1 \geq \eta_\epsilon(\mathbb{Z}^m) \cdot \|\widetilde{\mathbf{T_B}}\|$. In the following, we focus on the $\mathsf{SampleRight}_{\mathsf{non\text{-}sp}}$ algorithm. We define $\mathbf{R}' := \begin{bmatrix} -\mathbf{R} \\ \mathbf{I}\end{bmatrix} \in \mathbb{Z}^{2m\times m}$, $\mathbf{L} := [\mathbf{I}_{2m}|\mathbf{R}'] \in \mathbb{Z}^{2m\times 3m}$. Now we prove that $\mathbf{x} = \mathbf{L}\begin{bmatrix} \mathbf{p} \\ \mathbf{z}\end{bmatrix}$ is distributed statistically close to $D_{\mathbb{Z}^{2m},\sqrt{\Sigma}}$ where $\Sigma := \begin{bmatrix} \theta_1^2\mathbf{I} & \\ & \theta_2^2\mathbf{I}\end{bmatrix}$.

Note that $\mathbf{L}\begin{bmatrix} \Sigma_p & \\ & \zeta^2\mathbf{I}\end{bmatrix}\mathbf{L}^\top = \begin{bmatrix} \mathbf{I}_{2m} & \begin{bmatrix} -\mathbf{R} \\ \mathbf{I}\end{bmatrix}\end{bmatrix}\begin{bmatrix} \Sigma_p & \\ & \zeta^2\mathbf{I}\end{bmatrix}\begin{bmatrix} \mathbf{I}_{2m} \\ [-\mathbf{R}^\top \ \mathbf{I}]\end{bmatrix} = \begin{bmatrix} \theta_1^2\mathbf{I} & \\ & \theta_2^2\mathbf{I}\end{bmatrix}$. Let $\Lambda_\mathbf{L} := \mathbb{Z}^{3m} \cap \ker(\mathbf{L})$ that is an integer lattice. By Lemma 4, it suffices to show that $\eta_\epsilon(\Lambda_\mathbf{L}) \leq \sqrt{\begin{bmatrix} \Sigma_p & \\ & \zeta^2\mathbf{I}\end{bmatrix}}$. We prove this following the idea of [YJW23, Lemma 7]. Let $\mathbf{B_L} = \begin{bmatrix} -\mathbf{R}' \\ \mathbf{I}\end{bmatrix}$, then $\mathbf{B_L}$ is a basis of $\Lambda_\mathbf{L}$. The basis of $\Lambda_\mathbf{L}^*$ is

$$\mathbf{B}_\mathbf{L}^* = \mathbf{B_L}(\mathbf{B}_\mathbf{L}^\top\mathbf{B_L})^{-1} = \begin{bmatrix} -\mathbf{R}' \\ \mathbf{I}\end{bmatrix}\left(\begin{bmatrix} -\mathbf{R}'^\top & \mathbf{I}\end{bmatrix}\begin{bmatrix} -\mathbf{R}' \\ \mathbf{I}\end{bmatrix}\right)^{-1} = \begin{bmatrix} -\mathbf{R}' \\ \mathbf{I}\end{bmatrix}(\mathbf{R}'^\top\mathbf{R}' + \mathbf{I})^{-1}.$$

According to the definition of the smoothing parameter, we need to show

$$\sqrt{\begin{bmatrix} \Sigma_p & \\ & \zeta^2\mathbf{I}\end{bmatrix}} \geq \eta_\epsilon\big(\Lambda(\mathbf{B_L})\big),$$

i.e.,

$$(\mathbf{B}_\mathbf{L}^*)^\top\begin{bmatrix} \Sigma_p & \\ & \zeta^2\mathbf{I}\end{bmatrix}\mathbf{B}_\mathbf{L}^* \succ \eta_\epsilon^2(\mathbb{Z}^m).$$

This reduces to showing

$$(\mathbf{R}'^\top\mathbf{R}' + \mathbf{I})^{-\top}\begin{bmatrix} -\mathbf{R}'^\top & \mathbf{I}\end{bmatrix}\begin{bmatrix} \Sigma_p & \\ & \zeta^2\mathbf{I}\end{bmatrix}\begin{bmatrix} -\mathbf{R}' \\ \mathbf{I}\end{bmatrix}(\mathbf{R}'^\top\mathbf{R}' + \mathbf{I})^{-1} \succ \eta_\epsilon^2(\mathbb{Z}^m).$$

Now, substituting the definition $\mathbf{R}' := \begin{bmatrix} -\mathbf{R} \\ \mathbf{I}\end{bmatrix}$ (*this is a different step from [YJW23, Lemma 7]*), we can obtain

$$(\mathbf{R}^\top\mathbf{R}+2\mathbf{I})^{-\top}\left(\theta_1^2\mathbf{R}^\top\mathbf{R} - \zeta^2(\mathbf{R}^\top\mathbf{R})^2 - 2\zeta^2\mathbf{R}^\top\mathbf{R} + \theta_2^2\mathbf{I}\right)(\mathbf{R}^\top\mathbf{R}+2\mathbf{I})^{-1} \succ \eta_\epsilon^2(\mathbb{Z}^m). \tag{5}$$

Let $\mathbf{R}^\top\mathbf{R} = \mathbf{UVU}^{-1}$ be the eigenvalue decomposition where $\mathbf{V} = \mathrm{diag}(\lambda_1, \cdots, \lambda_m)$ with $\lambda_i$ be the eigenvalues. Eq. (5) can be rewritten as

$$(\mathbf{UVU}^{-1}+2\mathbf{I})^{-\top}\left(\theta_1^2\mathbf{UVU}^{-1} - \zeta^2\mathbf{UV}^2\mathbf{U}^{-1} - 2\zeta^2\mathbf{UVU}^{-1} + \theta_2^2\mathbf{I}\right)(\mathbf{UVU}^{-1}+2\mathbf{I})^{-1} \succ \eta_\epsilon^2(\mathbb{Z}^m),$$

i.e.,

$$\mathbf{U}(\mathbf{V}+2\mathbf{I})^{-\top}\Big(\theta_1^2\mathbf{V}-\zeta^2\mathbf{V}^2-2\zeta^2\mathbf{V}+\theta_2^2\mathbf{I}\Big)(\mathbf{V}+2\mathbf{I})^{-1}\mathbf{U}^{-1}\succ\eta_\epsilon^2(\mathbb{Z}^m).$$

Thus, we need to prove

$$\frac{\theta_1^2\lambda_i-\zeta^2\lambda_i^2-2\zeta^2\lambda_i+\theta_2^2}{(\lambda_i+2)^2}\geq\eta_\epsilon^2(\mathbb{Z}^m),$$

i.e.,

$$\theta_1^2\geq\big(\zeta^2+\eta_\epsilon^2(\mathbb{Z}^m)\big)\cdot\lambda_i+\big(4\eta_\epsilon^2(\mathbb{Z}^m)+2\zeta^2\big)+\frac{4\eta_\epsilon^2(\mathbb{Z}^m)-\theta_2^2}{\lambda_i}.$$

One can check that this condition is satisfied when $\theta_2\geq 2\eta_\epsilon(\mathbb{Z}^m)$ and $\theta_1^2\geq(\zeta^2+\eta_\epsilon^2(\mathbb{Z}^m))\|\mathbf{R}\|^2+2\zeta^2+4\eta_\epsilon^2(\mathbb{Z}^m)$.

By Schur complements, $\Sigma_p=\begin{bmatrix}\theta_1^2\mathbf{I}-\zeta^2\mathbf{R}\mathbf{R}^\top & \mathbf{R}\\ \mathbf{R}^\top & (\theta_2^2-\zeta^2)\mathbf{I}\end{bmatrix}$ is positive definite if and only if $\theta_2>c\zeta$ and $\theta_1>\sqrt{1+\frac{1}{c^2-1}}\zeta\|\mathbf{R}\|$. Let $c=\sqrt{2}$, then $\theta_2>\sqrt{2}\zeta$ and $\theta_1>\sqrt{2}\zeta\|\mathbf{R}\|$. This completes the proof. $\square$

## 5 Re-Randomization with Non-spherical Outputs

In this section, we first recall the re-randomization technique in [KY16], then design a new re-randomization algorithm with non-spherical outputs.

### 5.1 Recall re-randomization in [KY16] with spherical outputs

Katsumata and Yamada [KY16] proposed a re-randomization technique, which allows us to simulate the challenge ciphertext of IBE without resorting to the noise flooding technique in [Yam16]. More specifically, given a vector $\mathbf{b}+\mathbf{e}\in\mathbb{Z}_q^m$ and a matrix $\mathbf{R}\in\mathbb{Z}^{m\times m}$, they designed a re-randomization algorithm $\mathsf{ReRand}$[6] that outputs $\begin{bmatrix}\mathbf{I}\\ \mathbf{R}^\top\end{bmatrix}\mathbf{b}+\mathbf{e}_1$ where the error $\mathbf{e}_1$ is distributed statistically close to a discrete Gaussian distribution $D_{\mathbb{Z}^{2m},\sigma}$.

$\mathsf{ReRand}\big(\mathbf{R}\in\mathbb{Z}^{m\times m},\mathbf{b}+\mathbf{e}\in\mathbb{Z}_q^m,\delta,\sigma\in\mathbb{R}\big)$

1. sample $\mathbf{c}\leftarrow D_\delta^m$, sample $\mathbf{d}\leftarrow D_{\sqrt{\Sigma}}$ where $\Sigma:=\frac{\sigma^2}{2}\mathbf{I}-2\delta^2\begin{bmatrix}\mathbf{I}\\ \mathbf{R}^\top\end{bmatrix}\begin{bmatrix}\mathbf{I} & \mathbf{R}\end{bmatrix}$, and sample $\mathbf{f}\leftarrow D_{\mathbb{Z}^{2m}-\left(\begin{bmatrix}\mathbf{I}\\ \mathbf{R}^\top\end{bmatrix}\mathbf{c}+\mathbf{d}\right),\sigma/\sqrt{2}}$.
2. output $\begin{bmatrix}\mathbf{I}\\ \mathbf{R}^\top\end{bmatrix}\cdot\big((\mathbf{b}+\mathbf{e})+\mathbf{c}\big)+\mathbf{d}+\mathbf{f}$.

The parameters $\delta$ and $\sigma$ should satisfy $\delta\geq\eta_\epsilon(\mathbb{Z}^m)$ and $\sigma>2\delta\cdot\left\|\begin{bmatrix}\mathbf{I}\\ \mathbf{R}^\top\end{bmatrix}\right\|$.

[6] Note that the $\mathsf{ReRand}$ algorithm here is a special case of the one in [KY16], adapted to our use case.

### 5.2 Our new re-randomization with non-spherical outputs

To rerandomize the input and get a non-spherical error, i.e., a discrete Gaussian distribution $D_{\mathbb{Z}^{2m},\sqrt{\Sigma}}$ where $\Sigma := \begin{bmatrix} \sigma_1^2\mathbf{I} & \\ & \sigma_2^2\mathbf{I} \end{bmatrix}$ where $\sigma_1 \ll \sigma_2$, we design a new re-randomization algorithm $\mathsf{ReRand}_{\mathsf{non\text{-}sp}}$ as follows.

$\mathsf{ReRand}_{\mathsf{non\text{-}sp}}(\mathbf{R} \in \mathbb{Z}^{m\times m}, \mathbf{b}+\mathbf{e} \in \mathbb{Z}_q^m, \delta, \sigma_1, \sigma_2 \in \mathbb{R})$

1. sample $\mathbf{p} \leftarrow D_{\mathbb{Z}^{2m},\Sigma_p}$ where $\Sigma_p := \begin{bmatrix} \sigma_1^2\mathbf{I} & \\ & \sigma_2^2\mathbf{I} \end{bmatrix} - \delta^2 \begin{bmatrix} \mathbf{I} \\ \mathbf{R}^\top \end{bmatrix} [\,\mathbf{I}\ \mathbf{R}\,]$.
2. output $\begin{bmatrix} \mathbf{I} \\ \mathbf{R}^\top \end{bmatrix} \cdot (\mathbf{b}+\mathbf{e}) + \mathbf{p}$.

In the following theorem, we prove that our new re-randomization algorithm $\mathsf{ReRand}_{\mathsf{non\text{-}sp}}$ can re-randomize the error, making it distribution statistically close to a non-spherical Gaussian.

**Theorem 2.** *Let $q, m$ be positive integers and $\delta$ a positive real satisfying $\delta \geq \eta_\epsilon(\mathbb{Z}^m)$. Let $\mathbf{b} \in \mathbb{Z}_q^m$ be arbitrary and $\mathbf{e}$ chosen from $D_{\mathbb{Z}^m,\delta}$. Then for any $\mathbf{R} \in \mathbb{Z}^{m\times m}$, positive reals $\sigma_1 \geq 2\eta_\epsilon(\mathbb{Z}^m)$, $\sigma_1 > \sqrt{2}\delta$, $\sigma_2^2 \geq (\delta^2 + \eta_\epsilon^2(\mathbb{Z}^m))\|\mathbf{R}\|^2 + 2\delta^2 + 4\eta_\epsilon^2(\mathbb{Z}^m)$ and $\sigma_2 > \sqrt{2}\delta\|\mathbf{R}\|$, $\mathsf{ReRand}_{\mathsf{non\text{-}sp}}$ outputs $\begin{bmatrix} \mathbf{I} \\ \mathbf{R}^\top \end{bmatrix}\mathbf{b} + \mathbf{e}_1 \in \mathbb{Z}_q^{2m}$ where the error $\mathbf{e}_1$ is distributed statistically close to $D_{\mathbb{Z}^{2m},\sqrt{\Sigma}}$ where $\Sigma := \begin{bmatrix} \sigma_1^2\mathbf{I} & \\ & \sigma_2^2\mathbf{I} \end{bmatrix}$.*

*Proof.* The algorithm outputs the following,

$$\begin{bmatrix} \mathbf{I} \\ \mathbf{R}^\top \end{bmatrix}\mathbf{b} + \underbrace{\begin{bmatrix} \mathbf{I} \\ \mathbf{R}^\top \end{bmatrix}\mathbf{e} + \mathbf{p}}_{\mathbf{e}_1 := \text{“error”}} \in \mathbb{Z}_q^{2m}.$$

We analyze the error and show that it is distributed as in the statement. We define $\mathbf{R}' := \begin{bmatrix} \mathbf{I} \\ \mathbf{R}^\top \end{bmatrix} \in \mathbb{Z}^{2m\times m}$, $\mathbf{L} := [\mathbf{R}'|\mathbf{I}_{2m}] \in \mathbb{Z}^{2m\times 3m}$. Now we prove that $\mathbf{e}_1 = \mathbf{L}\begin{bmatrix} \mathbf{e} \\ \mathbf{p} \end{bmatrix}$ is distributed statistically close to $D_{\mathbb{Z}^{2m},\sqrt{\Sigma}}$ where $\Sigma := \begin{bmatrix} \sigma_1^2\mathbf{I} & \\ & \sigma_2^2\mathbf{I} \end{bmatrix}$.

Note that $\mathbf{L}\begin{bmatrix} \delta^2\mathbf{I} & \\ & \Sigma_p \end{bmatrix}\mathbf{L}^\top = \begin{bmatrix} \begin{bmatrix} \mathbf{I} \\ \mathbf{R}^\top \end{bmatrix} & \mathbf{I}_{2m} \end{bmatrix} \begin{bmatrix} \delta^2\mathbf{I} & \\ & \Sigma_p \end{bmatrix} \begin{bmatrix} [\,\mathbf{I}\ \mathbf{R}\,] \\ \mathbf{I}_{2m} \end{bmatrix} = \begin{bmatrix} \sigma_1^2\mathbf{I} & \\ & \sigma_2^2\mathbf{I} \end{bmatrix}$. Let $\Lambda_\mathbf{L} := \mathbb{Z}^{3m} \cap \ker(\mathbf{L})$ that is an integer lattice. By Lemma 4, it suffices to show that $\eta_\epsilon(\Lambda_\mathbf{L}) \leq \sqrt{\begin{bmatrix} \delta^2\mathbf{I} & \\ & \Sigma_p \end{bmatrix}}$. Let $\mathbf{B_L} = \begin{bmatrix} \mathbf{I} \\ -\mathbf{R}' \end{bmatrix}$, then $\mathbf{B_L}$ is a basis of $\Lambda_\mathbf{L}$. The dual basis of $\mathbf{B_L}$ is

$$\mathbf{B}_\mathbf{L}^* = \mathbf{B_L}(\mathbf{B}_\mathbf{L}^\top\mathbf{B_L})^{-1} = \begin{bmatrix} \mathbf{I} \\ -\mathbf{R}' \end{bmatrix}\left([\,\mathbf{I}\ -\mathbf{R}'^\top\,]\begin{bmatrix} \mathbf{I} \\ -\mathbf{R}' \end{bmatrix}\right)^{-1} = \begin{bmatrix} \mathbf{I} \\ -\mathbf{R}' \end{bmatrix}(\mathbf{I} + \mathbf{R}'^\top\mathbf{R}')^{-1}.$$

According to the definition of the smoothing parameter, we need to show

$$\sqrt{\begin{bmatrix} \delta^2\mathbf{I} & \\ & \Sigma_p \end{bmatrix}} \geq \eta_\epsilon(\Lambda(\mathbf{B_L})),$$

i.e.,

$$(\mathbf{B}_\mathbf{L}^*)^\top \begin{bmatrix} \delta^2\mathbf{I} & \\ & \Sigma_p \end{bmatrix} \mathbf{B}_\mathbf{L}^* \succ \eta_\epsilon^2(\mathbb{Z}^m).$$

This reduces to showing

$$(\mathbf{I}+\mathbf{R}'^{\top}\mathbf{R}')^{-\top}\left[\,\mathbf{I}\ -\mathbf{R}'^{\top}\right]\begin{bmatrix}\delta^2\mathbf{I} & \\ & \Sigma_p\end{bmatrix}\begin{bmatrix}\mathbf{I}\\ -\mathbf{R}'\end{bmatrix}(\mathbf{I}+\mathbf{R}'^{\top}\mathbf{R}')^{-1}\succ\eta_\epsilon^2(\mathbb{Z}^m).$$

Now, substituting the definition $\mathbf{R}' := \begin{bmatrix}\mathbf{I}\\ \mathbf{R}^{\top}\end{bmatrix}$, we can obtain

$$(2\mathbf{I}+\mathbf{R}\mathbf{R}^{\top})^{-\top}\left(\sigma_1^2\mathbf{I}+\sigma_2^2\mathbf{R}\mathbf{R}^{\top}-\delta^2(\mathbf{R}\mathbf{R}^{\top})^2-2\delta^2\mathbf{R}\mathbf{R}^{\top}\right)(2\mathbf{I}+\mathbf{R}\mathbf{R}^{\top})^{-1}\succ\eta_\epsilon^2(\mathbb{Z}^m). \tag{6}$$

Let $\mathbf{R}\mathbf{R}^{\top}=\mathbf{U}\mathbf{V}\mathbf{U}^{-1}$ be the eigenvalue decomposition where $\mathbf{V}=\mathrm{diag}(\lambda_1,\cdots,\lambda_m)$ with $\lambda_i$ be the eigenvalues. Eq. (6) can be rewritten as

$$(2\mathbf{I}+\mathbf{U}\mathbf{V}\mathbf{U}^{-1})^{-\top}\left(\sigma_1^2\mathbf{I}+\sigma_2^2\mathbf{U}\mathbf{V}\mathbf{U}^{-1}-\delta^2\mathbf{U}\mathbf{V}^2\mathbf{U}^{-1}-2\delta^2\mathbf{U}\mathbf{V}\mathbf{U}^{-1}\right)(2\mathbf{I}+\mathbf{U}\mathbf{V}\mathbf{U}^{-1})^{-1}\succ\eta_\epsilon^2(\mathbb{Z}^m),$$

i.e.,

$$\mathbf{U}(\mathbf{V}+2\mathbf{I})^{-\top}\left(\sigma_1^2\mathbf{I}+\sigma_2^2\mathbf{V}-\delta^2\mathbf{V}^2-2\delta^2\mathbf{V}\right)(\mathbf{V}+2\mathbf{I})^{-1}\mathbf{U}^{-1}\succ\eta_\epsilon^2(\mathbb{Z}^m).$$

Thus, we need to prove

$$\frac{\sigma_2^2\lambda_i-\delta^2\lambda_i^2-2\delta^2\lambda_i+\sigma_1^2}{(\lambda_i+2)^2}\geq\eta_\epsilon^2(\mathbb{Z}^m),$$

i.e.,

$$\sigma_2^2\geq\left(\eta_\epsilon^2(\mathbb{Z}^m)+\delta^2\right)\cdot\lambda_i+(4\eta_\epsilon^2(\mathbb{Z}^m)+2\delta^2)+\frac{4\eta_\epsilon^2(\mathbb{Z}^m)-\sigma_1^2}{\lambda_i}.$$

One can check that this condition is satisfied when $\sigma_1\geq 2\eta_\epsilon(\mathbb{Z}^m)$ and $\sigma_2^2\geq(\delta^2+\eta_\epsilon^2(\mathbb{Z}^m))\|\mathbf{R}\|^2+2\delta^2+4\eta_\epsilon^2(\mathbb{Z}^m)$.

By Schur complements, $\Sigma_p=\begin{bmatrix}(\sigma_1^2-\delta^2)\mathbf{I} & -\delta^2\mathbf{R}\\ -\delta^2\mathbf{R}^{\top} & \sigma_2^2\mathbf{I}-\delta^2\mathbf{R}^{\top}\mathbf{R}\end{bmatrix}$ is positive definite if and only if $\sigma_1>c\delta$ and $\sigma_2>\sqrt{1+\frac{1}{c^2-1}}\delta\|\mathbf{R}\|$. Let $c=\sqrt{2}$, then $\sigma_1>\sqrt{2}\delta$ and $\sigma_2>\sqrt{2}\delta\|\mathbf{R}\|$. This completes the proof. □

## 6 A General Framework of Lattice IBE with New Sampling and Re-Randomization Algorithms

In this section, we present a general framework of lattice IBE, incorporating our new sampling and re-randomization algorithms. The framework includes the IBE construction, its security proof, and its performance. Our framework is general in the sense that it is not restricted to any specific partition function, nor limited to integer or ring settings. We apply our framework to different IBEs and show the details in Sect. 7.

Our framework uses the following ingredients.

1. The trapdoor generation algorithm TrapGen.
2. A partitioning function $H$ with algorithms (PrtSmp, PubEval, TrapEval).
3. Parameters $n, m, q, t\in\mathbb{Z}, \delta, \theta_1, \theta_2, \sigma_1, \sigma_2\in\mathbb{R}$ which are determined by the security parameter $\lambda$ and/or the partitioning function $H$.
4. The sampling algorithms with non-spherical outputs $\mathsf{SampleLeft}_{\text{non-sp}}$ and $\mathsf{SampleRight}_{\text{non-sp}}$.
5. The re-randomization algorithm with non-spherical outputs $\mathsf{ReRand}_{\text{non-sp}}$.

### 6.1 Construction

We define the IBE scheme $(\mathsf{Setup}, \mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$ in the following.

$\mathsf{Setup}(1^\lambda)$**.** On inputs the security parameter $\lambda$, do:

1. generate a trapdoor pair $(\mathbf{B} \in \mathbb{Z}_q^{n\times m}, \mathbf{T_B} \in \mathbb{Z}^{m\times m}) \leftarrow \mathsf{TrapGen}(1^n, 1^m, q)$.
2. sample $\mathbf{u} \xleftarrow{\$} \mathbb{Z}_q^n$.
3. sample $\mathbf{C}_0 \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$, $\mathbf{C}_i \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$ for $i \in [t]$.
4. output $\mathsf{mpk} := \big(\mathbf{B}, \mathbf{u}, \mathbf{C}_0, \{\mathbf{C}_i\}_{i\in[t]}\big)$, keep $\mathsf{msk} := \mathbf{T_B}$.

$\mathsf{KeyGen}(\mathsf{mpk}, \mathsf{msk}, \mathsf{id})$**.** On inputs the master public key $\mathsf{mpk}$, the master secret key $\mathsf{msk}$ and an identity $\mathsf{id}$, do:

1. compute $\mathbf{C}_{\mathsf{id}} := \mathsf{PubEval}\big(\{\mathbf{C}_i\}_{i\in[t]}, \mathsf{id}\big) \in \mathbb{Z}_q^{n\times m}$.
2. sample $\mathbf{x} = \left[\begin{smallmatrix}\mathbf{x}_0\\ \mathbf{x}_1\end{smallmatrix}\right] \xleftarrow{\$} \mathsf{SampleLeft}_{\textsf{non-sp}}(\mathbf{B}, \mathbf{C}_0 + \mathbf{C}_{\mathsf{id}}, \mathbf{T_B}, \mathbf{u}, \theta_1, \theta_2)$ such that
$$[\mathbf{B}|\mathbf{C}_0 + \mathbf{C}_{\mathsf{id}}] \cdot \begin{bmatrix}\mathbf{x}_0\\ \mathbf{x}_1\end{bmatrix} = \mathbf{u} \bmod q, \text{ where } \mathbf{x}_0 \approx D_{\mathbb{Z}^m,\theta_1}, \mathbf{x}_1 \approx D_{\mathbb{Z}^m,\theta_2}.$$
3. output $\mathsf{sk}_{\mathsf{id}} := \mathbf{x} \in \mathbb{Z}^{2m}$.

$\mathsf{Enc}(\mathsf{mpk}, \mathsf{id}, \mu)$**.** On inputs the master public key $\mathsf{mpk}$, an identity $\mathsf{id}$ and a message $\mu \in \{0, 1\}$, do:

1. compute $\mathbf{C}_{\mathsf{id}} := \mathsf{PubEval}\big(\{\mathbf{C}_i\}_{i\in[t]}, \mathsf{id}\big) \in \mathbb{Z}_q^{n\times m}$.
2. sample $\mathbf{v} \xleftarrow{\$} \mathbb{Z}_q^n$, $e_0 \leftarrow D_{\mathbb{Z},\delta}$, $\mathbf{e}_1 \leftarrow D_{\mathbb{Z}^m,\sigma_1}$ and $\mathbf{e}_2 \leftarrow D_{\mathbb{Z}^m,\sigma_2}$.
3. compute $c_0 := \mathbf{v}^\top\mathbf{u} + e_0 + \lceil \frac{q}{2} \rceil \cdot \mu$, $\mathbf{c}_1^\top := \mathbf{v}^\top[\mathbf{B}|\mathbf{C}_0 + \mathbf{C}_{\mathsf{id}}] + [\mathbf{e}_1^\top|\mathbf{e}_2^\top]$.
4. output $\mathsf{ct} := (c_0, \mathbf{c}_1) \in \mathbb{Z}_q \times \mathbb{Z}_q^{2m}$.

$\mathsf{Dec}(\mathsf{sk}_{\mathsf{id}} := \mathbf{x}, \mathsf{ct} := (c_0, \mathbf{c}_1))$**.** On inputs the secret key $\mathsf{sk}_{\mathsf{id}}$ and a ciphertext, do:

1. output $\left\lfloor \frac{2}{q} \cdot (c_0 - \mathbf{c}_1^\top \cdot \mathbf{x}) \right\rceil \bmod 2$, where the rounding function $\lfloor\cdot\rceil$ is applied component-wise.

**Lemma 10 (Correctness).** *For any positive number $\omega$, if $\omega \cdot (\sigma_1\theta_1\sqrt{m} + \sigma_2\theta_2\sqrt{m} + \delta) \le q/5$, then the above IBE scheme has a decryption error that occurs with probability at most $6e^{-\pi\omega^2} + 2^{-m+1}$.*

*Proof.* For the $\mathsf{Dec}$ algorithm, we show that the error term in decryption would not exceed $q/5$. Specifically, the decryption algorithm calculates

$$\begin{aligned} c_0 - \mathbf{c}_1^\top \cdot \mathbf{x} &= \mathbf{v}^\top\mathbf{u} + e_0 + \left\lceil \frac{q}{2} \right\rceil \cdot \mu - (\mathbf{v}^\top \mathsf{pk}_{\mathsf{id}} + [\mathbf{e}_1^\top|\mathbf{e}_2^\top]) \cdot \mathbf{x} \\ &= \left\lceil \frac{q}{2} \right\rceil \cdot \mu + \underbrace{e_0 - \left\langle \begin{bmatrix}\mathbf{e}_1\\ \mathbf{e}_2\end{bmatrix}, \begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix} \right\rangle}_{:=\text{error term}}. \end{aligned}$$

Then we analyze the error term:

1. By $\mathbf{e}_1 \leftarrow D_{\mathbb{Z}^m,\sigma_1}$ and Lemma 2, we have $\Pr[|\langle \mathbf{e}_1, \mathbf{x}_1\rangle| \ge t_1] \le 2e^{-\pi\cdot\frac{t_1^2}{\|\mathbf{x}_1\|^2\sigma_1^2}}$. By $\mathbf{x}_1 \leftarrow D_{\mathbb{Z}^m,\theta_1}$ and Lemma 3, we have $\Pr[\|\mathbf{x}_1\| \ge \theta_1\sqrt{m}] \le 2^{-m}$.

2. By $\mathbf{e}_2 \leftarrow D_{\mathbb{Z}^m,\sigma_2}$ and Lemma 2, we have $\Pr[|\langle \mathbf{e}_2, \mathbf{x}_2\rangle| \geq t_2] \leq 2e^{-\pi \cdot \frac{t_2^2}{\|\mathbf{x}_2\|^2 \sigma_2^2}}$. By $\mathbf{x}_2 \leftarrow D_{\mathbb{Z}^m,\theta_2}$ and Lemma 3, we have $\Pr[\|\mathbf{x}_2\| \geq \theta_2\sqrt{m}] \leq 2^{-m}$.
3. By $e_0 \leftarrow D_{\mathbb{Z},\delta}$ and Lemma 1, we have $\Pr[|e_0| \geq t_3] \leq 2e^{-\pi \cdot \frac{t_3^2}{\delta^2}}$.

Taking $t_1 = \omega\sigma_1\theta_1\sqrt{m}$, $t_2 = \omega\sigma_2\theta_2\sqrt{m}$ and $t_3 = \omega\delta$, then

$$\Pr\left[\left|e_0 - \left\langle \begin{bmatrix}\mathbf{e}_1\\ \mathbf{e}_2\end{bmatrix}, \begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}\right\rangle\right| \geq t_1 + t_2 + t_3\right] \leq 6e^{-\pi\omega^2} + 2^{-m+1}.$$

Therefore, if $\omega \cdot (\sigma_1\theta_1\sqrt{m} + \sigma_2\theta_2\sqrt{m} + \delta) \leq q/5$, then the decryption error occurs with probability at most $6e^{-\pi\omega^2} + 2^{-m+1}$. This completes the proof. $\square$

### 6.2 Security Proof

We prove the following theorem which shows the security of our IBE scheme. Our proof follows the idea of [Yam17].

**Theorem 3.** *If $H : \mathcal{K} \times \mathcal{ID} \to \mathcal{T}$ is a partitioning function with $\Delta$-expanding algorithms* (PubEval, TrapEval)*, our IBE scheme achieves adaptively-anonymous security under the* $\mathsf{LWE}_{n,m+1,q,\delta}$ *assumption.*

*Proof.* Let $\mathcal{A}$ be a PPT adversary that breaks the adaptively-anonymous security of the scheme. In addition, let $\epsilon = \epsilon(\lambda)$ and $Q = Q(\lambda)$ be its advantage and the upper bound of the number of key extraction queries, respectively. By assumption, $Q(\lambda)$ is polynomially bounded and there exists a noticeable function $\epsilon_0(\lambda)$ such that $\epsilon(\lambda) \geq \epsilon_0(\lambda)$ holds for infinitely many $\lambda$. By the property of the partitioning function (Def. 3, Item 1), we have that

$$\boldsymbol{\kappa} \in \mathcal{K} \quad \text{where} \quad \boldsymbol{\kappa} \xleftarrow{\$} \mathsf{PrtSmp}(1^\lambda, Q, \epsilon_0)$$

holds with probability 1 for all sufficiently large $\lambda$. Here, PrtSmp is the sampling algorithm associated with $H$. In the following, we prove the security of the scheme via a sequence of games. Each game defines a value $\mathsf{coin}' \in \{0,1\}$. We define $E_i$ to be the event that $\mathsf{coin}' = \mathsf{coin}$.

*Game 0.* This is the real game. At the end of the game, $\mathcal{A}$ outputs a guess $\widehat{\mathsf{coin}}$ for coin. Finally, the challenger sets $\mathsf{coin}' = \widehat{\mathsf{coin}}$. By definition, we have

$$\left|\Pr[E_0] - \frac{1}{2}\right| = \left|\Pr[\mathsf{coin}' = \mathsf{coin}] - \frac{1}{2}\right| = \left|\Pr[\widehat{\mathsf{coin}} = \mathsf{coin}] - \frac{1}{2}\right| = \epsilon.$$

*Game 1.* In this game, the challenger performs an additional abort check at the end of the game. In the beginning of the game, the challenger runs $\boldsymbol{\kappa} \xleftarrow{\$} \mathsf{PrtSmp}(1^\lambda, Q, \epsilon_0)$. At the end, the challenger checks whether the following condition holds:

$$H(\boldsymbol{\kappa}, \mathsf{id}^{(1)}) \neq 0 \,\wedge\, \cdots \,\wedge\, H(\boldsymbol{\kappa}, \mathsf{id}^{(Q)}) \neq 0 \,\wedge\, H(\boldsymbol{\kappa}, \mathsf{id}^*) = 0,$$

where $\mathsf{id}^*$ is the challenge identity, and $\mathsf{id}^{(1)}, \cdots, \mathsf{id}^{(Q)}$ are identities for which $\mathcal{A}$ has made key extraction queries. If it does not hold, the challenger ignores the output $\widehat{\mathsf{coin}}$ of $\mathcal{A}$, and sets $\mathsf{coin}' \xleftarrow{\$} \{0,1\}$. In this case, we say that the challenger aborts. If the condition holds, the challenger sets $\mathsf{coin}' = \widehat{\mathsf{coin}}$. By the second property of the partitioning function (Def. 3, Item 2) and Lemma 9,

$$\left|\Pr[E_1] - \frac{1}{2}\right| \geq \gamma_{\min}\epsilon - \frac{1}{2}(\gamma_{\max} - \gamma_{\min}) \geq \gamma_{\min} \cdot \epsilon_0 - \frac{1}{2}(\gamma_{\max} - \gamma_{\min}) = \tau,$$

holds for infinitely many $\lambda$ and a noticeable function $\tau = \tau(\lambda)$, where $\gamma_{\min}$, $\gamma_{\max}$ and $\tau$ are specified by $\epsilon_0$, $Q$, and the underlying partitioning function $H$.

*Game 2.* In this game, we change the way that the master public key $\left(\mathbf{C}_0, \{\mathbf{C}_i\}_{i\in[t]}\right)$ is generated.

– In Game 1, $\mathbf{C}_0 \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$ and $\mathbf{C}_i \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$ for $i \in [t]$.
– In Game 2, the challenger runs $\boldsymbol{\kappa} \xleftarrow{\$} \mathsf{PrtSmp}(1^\lambda, Q, \epsilon_0)$, splits $\boldsymbol{\kappa} \coloneqq (\kappa_1, \cdots, \kappa_t)$, and samples $\mathbf{R}_0, \mathbf{R}_i \xleftarrow{\$} \{-1,1\}^{m\times m}$ for $i \in [t]$. Then, it defines $\mathbf{C}_0 = \mathbf{B}\mathbf{R}_0$, $\mathbf{C}_i = \mathbf{B}\mathbf{R}_i + \kappa_i\mathbf{G}$ for $i \in [t]$.

By the Leftover Hash Lemma (Lemma 8) and a simple hybrid argument, the following two distributions are statistically close:

$$\left(\mathbf{B}, \mathbf{B}\mathbf{R}_0, \{\mathbf{B}\mathbf{R}_i + \kappa_i\mathbf{G}\}_{i\in[t]}\right) \approx \left(\mathbf{B}, \mathbf{C}_0, \{\mathbf{C}_i\}_{i\in[t]}\right),$$

where $\mathbf{C}_0, \mathbf{C}_i \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$. Thus, $|\Pr[E_1] - \Pr[E_2]| = \mathsf{negl}(n)$.

*Game 3.* In this game, we change the generation of $\mathbf{B}$ in the master public key and the generation of user's secret key $\mathsf{sk}_{\mathsf{id}}$.

– In Game 2, $\mathbf{B}$ is generated by the $\mathsf{TrapGen}$ algorithm along with a corresponding trapdoor $\mathbf{T_B}$, $\mathsf{sk}_{\mathsf{id}}$ is sampled by the $\mathsf{SampleLeft}_{\mathsf{non\text{-}sp}}$ algorithm, i.e.,

$$\mathsf{sk}_{\mathsf{id}} \xleftarrow{\$} \mathsf{SampleLeft}_{\mathsf{non\text{-}sp}}(\mathbf{B}, \mathbf{C}_0 + \mathbf{C}_{\mathsf{id}}, \mathbf{T_B}, \mathbf{u}, \theta_1, \theta_2).$$

– In Game 3, $\mathbf{B}$ is chosen uniformly at random over $\mathbb{Z}_q^{n\times m}$ and $\mathsf{sk}_{\mathsf{id}}$ is sampled by the $\mathsf{SampleRight}_{\mathsf{non\text{-}sp}}$ algorithm. Specifically, the challenger computes $H(\boldsymbol{\kappa}, \mathsf{id})$ and

$$\mathbf{R}_{\mathsf{id}} \coloneqq \mathsf{TrapEval}(\mathbf{B}, \{\mathbf{R}_i\}_{i\in[t]}, \{\kappa_i\}_{i\in[t]}, \mathsf{id}),$$

i.e., $\mathbf{C}_{\mathsf{id}} = \mathbf{B}\mathbf{R}_{\mathsf{id}} + H(\boldsymbol{\kappa}, \mathsf{id})\mathbf{G}$. Then, it samples $\mathsf{sk}_{\mathsf{id}}$ as follows

$$\mathsf{sk}_{\mathsf{id}} \xleftarrow{\$} \mathsf{SampleRight}_{\mathsf{non\text{-}sp}}(\mathbf{B}, \mathbf{G}, \mathbf{T_G}, \mathbf{R}_0 + \mathbf{R}_{\mathsf{id}}, H(\boldsymbol{\kappa}, \mathsf{id}), \mathbf{u}, \theta_1, \theta_2, \zeta)$$

By Lemma 5 and Theorem 1, we have $|\Pr[E_2] - \Pr[E_3]| = \mathsf{negl}(n)$.

*Game 4.* In this game, we change the way that the challenge ciphertext $\mathbf{c}_1^*$ is generated when $\mathsf{coin} = 0$.

– In Game 3, $\mathbf{c}_1^* := \mathsf{pk}_{\mathsf{id}^*}^\top \mathbf{v} + \left[\begin{smallmatrix}\mathbf{e}_1\\ \mathbf{e}_2\end{smallmatrix}\right]$, where $\mathbf{v} \xleftarrow{\$} \mathbb{Z}_q^n$, $\mathbf{e}_1 \leftarrow D_{\mathbb{Z}^m,\sigma_1}$ and $\mathbf{e}_2 \leftarrow D_{\mathbb{Z}^m,\sigma_2}$. Note that $\mathsf{pk}_{\mathsf{id}^*} := [\mathbf{B}|\mathbf{B}(\mathbf{R}_0 + \mathbf{R}_{\mathsf{id}^*})]$ since $H(\boldsymbol{\kappa}, \mathsf{id}^*) = 0$, then
$$\mathbf{c}_1^* := \begin{bmatrix}\mathbf{B}^\top \\ (\mathbf{R}_0^\top + \mathbf{R}_{\mathsf{id}^*}^\top)\mathbf{B}^\top\end{bmatrix}\mathbf{v} + \begin{bmatrix}\mathbf{e}_1\\ \mathbf{e}_2\end{bmatrix}.$$
– In Game 4, the challenger samples $\mathbf{e} \leftarrow D_{\mathbb{Z}^m,\delta}$, and defines
$$\mathbf{c}_1^* := \mathsf{ReRand}_{\mathsf{non\text{-}sp}}(\mathbf{R}_0 + \mathbf{R}_{\mathsf{id}^*}, \mathbf{B}^\top\mathbf{v} + \mathbf{e}, \delta, \sigma_1, \sigma_2).$$

By Theorem 2, we have that $\mathsf{ReRand}_{\mathsf{non\text{-}sp}}$ outputs $\left[\begin{smallmatrix}\mathbf{I}\\ \mathbf{R}_0^\top + \mathbf{R}_{\mathsf{id}^*}^\top\end{smallmatrix}\right]\mathbf{B}^\top\mathbf{v} + \left[\begin{smallmatrix}\mathbf{e}_1\\ \mathbf{e}_2\end{smallmatrix}\right]$ where $\left[\begin{smallmatrix}\mathbf{e}_1\\ \mathbf{e}_2\end{smallmatrix}\right]$ is distributed statistically close to $D_{\mathbb{Z}^{2m},\sqrt{\Sigma}}$ where $\Sigma := \left[\begin{smallmatrix}\sigma_1^2\mathbf{I} & \\ & \sigma_2^2\mathbf{I}\end{smallmatrix}\right]$. Thus, $|\Pr[E_3] - \Pr[E_4]| = \mathsf{negl}(n)$.

*Game 5.* In this game, we further change the way that the challenge ciphertext $(c_0^*, \mathbf{c}_1^*)$ is generated when $\mathsf{coin} = 0$.

– In Game 4, the challenger samples $e_0 \leftarrow D_{\mathbb{Z},\delta}$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m,\delta}$, and defines
$$c_0^* := \mathbf{v}^\top\mathbf{u} + e_0 + \left\lceil \frac{q}{2} \right\rceil \cdot \mu, \quad \mathbf{c}_1^* := \mathsf{ReRand}_{\mathsf{non\text{-}sp}}(\mathbf{R}_0 + \mathbf{R}_{\mathsf{id}^*}, \mathbf{B}^\top\mathbf{v} + \mathbf{e}, \delta, \sigma_1, \sigma_2).$$
– In Game 5, the challenger samples $y_0 \xleftarrow{\$} \mathbb{Z}_q$, $\mathbf{y} \xleftarrow{\$} \mathbb{Z}_q^m$, $e_0 \leftarrow D_{\mathbb{Z},\delta}$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m,\delta}$, and defines
$$c_0^* := y_0 + e_0 + \left\lceil \frac{q}{2} \right\rceil \cdot \mu, \quad \mathbf{c}_1^* := \mathsf{ReRand}_{\mathsf{non\text{-}sp}}(\mathbf{R}_0 + \mathbf{R}_{\mathsf{id}^*}, \mathbf{y} + \mathbf{e}, \delta, \sigma_1, \sigma_2).$$

We claim that $|\Pr[E_4] - \Pr[E_5]| = \mathsf{negl}(n)$ assuming the $\mathsf{LWE}_{n,m+1,q,\delta}$ assumption (Def. 2) holds. To show this, we define an adversary $\mathcal{B}$ against the LWE problem as follows.

$\mathcal{B}$ is given the LWE instance $(\mathbf{B}', \mathbf{w}' = \mathbf{y}' + \mathbf{e}') \in \mathbb{Z}_q^{n\times(m+1)} \times \mathbb{Z}_q^{m+1}$ where $\mathbf{e}' \leftarrow D_{\mathbb{Z}^{m+1},\delta}$. The task of $\mathcal{B}$ is to distinguish whether $\mathbf{y}'^\top = \mathbf{v}^\top\mathbf{B}'$ for $\mathbf{v} \xleftarrow{\$} \mathbb{Z}_q^n$ or $\mathbf{y}' \xleftarrow{\$} \mathbb{Z}_q^{m+1}$.

**Setup.** $\mathcal{B}$ sets $\mathbf{u}$ as the first column of $\mathbf{B}'$, $\mathbf{B}$ as the last $m$ columns of $\mathbf{B}'$, and sets $(\mathbf{C}_0, \{\mathbf{C}_i\}_{i\in[t]})$ as in Game 2, outputs $\mathsf{mpk} := (\mathbf{B}, \mathbf{u}, \mathbf{C}_0, \{\mathbf{C}_i\}_{i\in[t]})$.

**Phase 1 & Phase 2.** When $\mathcal{A}$ makes key extraction queries, $\mathcal{B}$ responds as in Game 3 without knowing $\mathbf{T_B}$.

**Challenge.** When $\mathcal{A}$ makes the challenge query for the identity $\mathsf{id}^*$ and a message $m$, $\mathcal{B}$ first picks $\mathsf{coin} \xleftarrow{\$} \{0,1\}$. If $\mathsf{coin} = 0$, $\mathcal{B}$ generates the challenge ciphertext as
$$c_0^* := w_0 + \left\lceil \frac{q}{2} \right\rceil \cdot \mu, \quad \mathbf{c}_1^* := \mathsf{ReRand}_{\mathsf{non\text{-}sp}}(\mathbf{R}_0 + \mathbf{R}_{\mathsf{id}^*}, \mathbf{w}, \delta, \sigma_1, \sigma_2),$$
where $w_0$ is the first element of $\mathbf{w}'$ and $\mathbf{w}$ is the last $m$ elements of $\mathbf{w}'$, and returns it to $\mathcal{A}$. If $\mathsf{coin} = 1$, $\mathcal{B}$ returns a random ciphertext.

$\mathsf{Guess.}$ $\mathcal{A}$ outputs its guess $\mathsf{coin}'$. $\mathcal{B}$ outputs 1 if $\mathsf{coin}' = \mathsf{coin}$ and 0 otherwise.

It can be seen that if $(\mathbf{B}', \mathbf{w}')$ is a valid LWE sample (i.e., $\mathbf{y}'^{\top} = \mathbf{v}^{\top}\mathbf{B}'$), the view of $\mathcal{A}$ corresponds to Game 4. If $\mathbf{v}' \xleftarrow{\$} \mathbb{Z}_q^{m+1}$, the view of $\mathcal{A}$ corresponds to Game 5. Thus, the advantage of $\mathcal{B}$ is $|\Pr[E_4] - \Pr[E_5]|$. By the $\mathsf{LWE}_{n,m+1,q,\delta}$ assumption, we have $|\Pr[E_4] - \Pr[E_5]| = \mathsf{negl}(n)$.

*Game 6.* In this game, we further change the way that the challenge is generated when $\mathsf{coin} = 0$.

– In Game 5, the challenger samples $\mathbf{y} \xleftarrow{\$} \mathbb{Z}_q^m$, $\mathbf{e} \leftarrow D_{\mathbb{Z}^m,\delta}$, and defines

$$\mathbf{c}_1^* \coloneqq \mathsf{ReRand}_{\mathsf{non\text{-}sp}}(\mathbf{R}_0 + \mathbf{R}_{\mathsf{id}^*}, \mathbf{y} + \mathbf{e}, \delta, \sigma_1, \sigma_2).$$

– In Game 6, the challenger samples $\mathbf{e}_1 \leftarrow D_{\mathbb{Z}^m,\sigma_1}$, $\mathbf{e}_2 \leftarrow D_{\mathbb{Z}^m,\sigma_2}$, and defines

$$\mathbf{c}_1^* \coloneqq \begin{bmatrix} \mathbf{I} \\ \mathbf{R}_0^{\top} + \mathbf{R}_{\mathsf{id}^*}^{\top} \end{bmatrix} \mathbf{y} + \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \end{bmatrix}.$$

By Theorem 2, we have that $\mathsf{ReRand}_{\mathsf{non\text{-}sp}}$ outputs $\begin{bmatrix} \mathbf{I} \\ \mathbf{R}_0^{\top} + \mathbf{R}_{\mathsf{id}^*}^{\top} \end{bmatrix}\mathbf{y} + \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \end{bmatrix}$ where $\begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \end{bmatrix}$ is distributed statistically close to $D_{\mathbb{Z}^{2m},\sqrt{\Sigma}}$ where $\Sigma \coloneqq \begin{bmatrix} \sigma_1^2\mathbf{I} & \\ & \sigma_2^2\mathbf{I} \end{bmatrix}$. Thus, $|\Pr[E_5] - \Pr[E_6]| = \mathsf{negl}(n)$.

*Game 7.* In this game, we change the challenge ciphertext to be a random vector, regardless of whether $\mathsf{coin} = 0$ or $\mathsf{coin} = 1$. In this game, the value $\mathsf{coin}$ is independent of the view of $\mathcal{A}$. Therefore, $\Pr[E_7] = \frac{1}{2}$.

We now proceed to bound $|\Pr[E_6] - \Pr[E_7]|$. Since Game 6 and Game 7 differ only in the creation of the challenge ciphertext when $\mathsf{coin} = 0$, we focus on this case. By the Leftover Hash Lemma (Lemma 8), the following distributions are statistically close:

$$(\mathbf{B}, \mathbf{B}\mathbf{R}_0, \mathbf{y}^{\top}, \mathbf{y}^{\top}\mathbf{R}_0) \approx (\mathbf{B}, \mathbf{B}', \mathbf{y}^{\top}, \mathbf{y}'^{\top}) \approx (\mathbf{B}, \mathbf{B}\mathbf{R}_0, \mathbf{y}^{\top}, \mathbf{y}'^{\top}), \tag{7}$$

where $\mathbf{B}, \mathbf{B}' \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$, $\mathbf{R}_0 \xleftarrow{\$} \{-1,1\}^{m\times m}$, $\mathbf{y}, \mathbf{y}' \xleftarrow{\$} \mathbb{Z}_q^m$. It can be seen that the first and the second distributions are statistically close, by applying Lemma 8 for $\begin{bmatrix} \mathbf{B} \\ \mathbf{y}^{\top} \end{bmatrix} \in \mathbb{Z}_q^{(n+1)\times m}$ and $\mathbf{R}_0$. It can also be seen that the second and the third distributions are statistically close, by applying the same lemma for $\mathbf{B}$ and $\mathbf{R}_0$. From the above, we have that the following distributions are statistically close:

$$\begin{aligned}(\mathbf{B}, \mathbf{B}\mathbf{R}_0, \mathbf{y}^{\top} + \mathbf{e}_1^{\top}, \mathbf{y}^{\top}(\mathbf{R}_0 + \mathbf{R}_{\mathsf{id}^*}) + \mathbf{e}_2^{\top}) &\approx (\mathbf{B}, \mathbf{R}_0, \mathbf{y}^{\top} + \mathbf{e}_1^{\top}, \mathbf{y}'^{\top} + \mathbf{y}^{\top}\mathbf{R}_{\mathsf{id}^*} + \mathbf{e}_2^{\top}) \\ &\approx (\mathbf{B}, \mathbf{B}\mathbf{R}_0, \mathbf{y}^{\top}, \mathbf{y}'^{\top}),\end{aligned}$$

where $\mathbf{B} \xleftarrow{\$} \mathbb{Z}_q^{n\times m}$, $\mathbf{R}_0 \xleftarrow{\$} \{-1,1\}^{m\times m}$, $\mathbf{y}, \mathbf{y}' \xleftarrow{\$} \mathbb{Z}_q^m$. The first and the second distributions above are statistically close by Eq. (7), whereas the second and the third distributions are statistically close by the fact that $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{R}_i$, which are used to compute $\mathbf{R}_{\mathsf{id}^*}$, are chosen independently random from other variables. Therefore, we conclude that $|\Pr[E_6] - \Pr[E_7]| = \mathsf{negl}(n)$.

**Analysis.** From the above, we have that

$$\begin{aligned}\left|\Pr[E_7] - \frac{1}{2}\right| &= \left|\Pr[E_1] - \frac{1}{2} + \sum_{i=1}^{6}\big(\Pr[E_{i+1}] - \Pr[E_i]\big)\right| \\ &\geq \left|\Pr[E_1] - \frac{1}{2}\right| - \sum_{i=1}^{6}|\Pr[E_{i+1}] - \Pr[E_i]| \\ &\geq \tau(\lambda) - \mathsf{negl}(\lambda)\end{aligned}$$

for infinitely many $\lambda$. Since $\Pr[E_7] = \frac{1}{2}$, we have $\tau(\lambda) \leq \mathsf{negl}(\lambda)$ for infinitely many $\lambda$. This is a contradiction since $\tau(\lambda)$ is noticeable. This completes the proof. $\square$

### 6.3 Performance of the resulting IBE

In this subsection, we show the master public key size, the LWE parameter and the reduction cost of the resulting IBE with respect to a partitioning function $H$. Note that, we use the same measurement for reduction cost as in [BR09].

**Theorem 4.** *Let $H$ be a partitioning function (Def. 3) with partitioning key $\boldsymbol{\kappa} = (\kappa_1, \cdots, \kappa_t)$ and the quality $\tau$, and $(\mathsf{PubEval}, \mathsf{TrapEval})$ be $\Delta$-expanding algorithms (Def. 4) with respect to $H$. Let $\lambda$ be the security parameter, $n$ be an integer such that $n = \Theta(\lambda)$, $\theta_1, \theta_2$ be two positive reals that satisfy the condition in Theorem 1, $\sigma_1, \sigma_2$ be two positive reals that satisfy the condition in Theorem 2, $\delta$ be a positive real such that $= O(\sqrt{n})$, $q$ be an integer that satisfies the condition in Lemma 10, $m$ be an positive integer such that $m = O(n \log q)$. Then, the performance of our IBE framework can be quantified as follows:*

$$|\mathsf{mpk}|\# = \Theta(t), \quad \text{LWE param } \frac{1}{\alpha} = \Delta \cdot \tilde{O}(n), \quad \text{Reduction cost} = \tau.$$

*Proof.* It is clear that $\mathsf{mpk}$ contains $t+2$ basic matrices ($\mathbb{Z}_q^{n\times m}$). In the proof of Theorem 3, there is a reduction to LWE to bound the difference between Game 4 and 5. All the other games are statistically close except for Game 0 and 1. So the reduction cost comes from the additional abort check in Game 1. According to [BR09], the reduction cost of this abort check is exactly $\tau$. So the resulting IBE has a reduction cost $\tau$.

To calculate the LWE parameter $\frac{1}{\alpha} = \frac{q}{\delta}$ where $\delta$ is the Gaussian width of the error in LWE samples (as in Def. 2), we first consider the modulus $q$. By Def. 4, if the algorithms $(\mathsf{PubEval}, \mathsf{TrapEval})$ are $\Delta$-expanding respect to $H$, then we have that $\|\mathbf{R}_{\mathsf{id}}\| \leq \Delta \cdot \max_{i\in[t]} \|\mathbf{R}_i\|$. Since $\mathbf{R}_i \in \{-1, 1\}^{m\times m}$, we have

$\|\mathbf{R}_i\| = \sqrt{m} = \tilde{O}(n^{0.5})$. Thus, $\|\mathbf{R}_{\mathsf{id}}\| \le \Delta \cdot \tilde{O}(n^{0.5})$. By Theorem 1, Theorem 2 and Lemma 10, if we choose $\omega = \omega(\sqrt{\log n})$ in Lemma 10, then we have

$$q = \tilde{O}(\theta_1\sigma_1 + \theta_2\sigma_2) \cdot \tilde{O}(n^{0.5}) = \tilde{O}(\Delta \cdot n^{0.5} \cdot \delta) \cdot \tilde{O}(n^{0.5}) = \Delta \cdot \tilde{O}(n) \cdot \delta.$$

Thus, the LWE parameter $\frac{1}{\alpha} = \frac{q}{\delta} = \Delta \cdot \tilde{O}(n)$. □

*Remark 1.* Note that our new framework improves the ABB framework that necessitates LWE parameter $\frac{1}{\alpha} = \Delta^2 \cdot \tilde{O}(n^{1.5})$.

## 7 Application to IBEs

In this section, we apply our new lattice-based IBE framework to existing IBE schemes. Specifically, we focus on three representative IBEs: the Agrawal-Boneh-Boyen (ABB) IBE [ABB10], an original construction for adaptively secure lattice IBE in the standard model; the Yamada IBE [Yam17], which introduces the notion of partitioning function; and the Abla-Liu-Wang-Wang (ALWW) IBE [ALWW21], which achieves the smallest asymptotic master public key size among known IBE schemes. These schemes cover different partitioning techniques, including both integer and ring variants. Beyond these cases, our framework can be readily adapted to many other IBE constructions. See Tab. 2 for a detailed comparison.

### 7.1 Application to the ABB IBE

Here, we apply our framework to the ABB IBE [ABB10]. Specifically, Agrawal et el. [ABB10] first used the Waters' hash [Wat05] as their partitioning function, but the scheme requires the modulus $q$ to be exponential, leading to a large inefficiency. To this end, Boyen [Boy10] designed a partitioning function more suited to the algebraic constraints of lattice.

**Definition 5 ([ABB10, Boy10]).** *Let $n, k, q$ be positive integers such that $n = \Theta(\lambda)$, $k \mid n$ and $q$ is prime. Let $\ell$ be the length of the identity such that $\ell = O(n)$. Let $H^{\mathsf{frd}}$ be a function, which takes vector in $\mathbb{Z}_q^k$ with $k \mid n$ and outputs a block-diagonal matrix in size $\mathbb{Z}_q^{n\times n}$. Let the key space $\mathcal{K} := (\mathbb{Z}_q^k)^\ell$, the identity $\mathcal{ID} := \{0,1\}^\ell$, the range $\mathcal{T} := \mathbb{Z}_q^{n\times n}$. Then, for a key $\boldsymbol{\kappa} := \{\kappa_i\}_{i\in[\ell]} \in \mathcal{K}$ where $\kappa_i \in \mathbb{Z}_q^k$, an identity $\mathsf{id} := (\mathsf{id}_1, \cdots, \mathsf{id}_\ell) \in \mathcal{ID}$ where $\mathsf{id}_i$ is the $i$-th bit of an identity $\mathsf{id}$, the function $H$ is defined as*

$$H\left(\boldsymbol{\kappa} := \{\kappa_i\}_{i\in[\ell]}, \mathsf{id}\right) := \mathbf{I}_n + \sum_{i=1}^{\ell} (-1)^{\mathsf{id}_i} \cdot H^{\mathsf{frd}}(\kappa_i).$$

The following lemma characterizes important properties of the function $H$.

**Lemma 11 ([ABB10, Boy10]).** *The function $H$ defined above is a partitioning function (Def. 3). More specifically, there exists an algorithm $\mathsf{PrtSmp}_{\mathsf{ABB}}$*

*satisfying the requirement of [Def. 3, Item 1], and the quality of the partitioning function is* $\tau := O\left(\frac{\epsilon^3}{Q^2}\right)$*, which is noticeable as required in [Def. 3, Item 2]. Furthermore, there exists two deterministic algorithms* ($\mathsf{PubEval}_{\mathsf{ABB}}, \mathsf{TrapEval}_{\mathsf{ABB}}$) *are* $\Delta$*-expanding for the function family* $\{H\}$ *(Def. 4), where* $\Delta := \ell$*.*

The performance of the original ABB IBE can be quantified as follows.

$$|\mathsf{mpk}|\# = O(\lambda), \quad \textit{LWE param } \frac{1}{\alpha} = \tilde{O}(n^{3.5}), \quad \textit{reduction cost} = O\left(\frac{\epsilon^3}{Q^2}\right).$$

The following corollary quantifies the performance of IBE which combines the original ABB IBE and our new framework.

**Corollary 1.** *If we plug the partitioning function* $H$ *in Def. 5 and the corresponding algorithms* ($\mathsf{PrtSmp}_{\mathsf{ABB}}, \mathsf{PubEval}_{\mathsf{ABB}}, \mathsf{TrapEval}_{\mathsf{ABB}}$) *(in Lemma 11) into the framework in Sect. 6.1, the resulting IBE scheme satisfies that*

$$|\mathsf{mpk}|\# = O(\lambda), \quad \text{LWE param } \frac{1}{\alpha} = \tilde{O}(n^2), \quad \text{reduction cost} = O\left(\frac{\epsilon^3}{Q^2}\right).$$

*Proof.* This result follows directly from Theorem 4 and Lemma 11. □

### 7.2 Application to the Yamada IBE

Here, we apply our framework to the Yamada IBE [Yam17]. Specifically, Yamada modified both the admissible hash [BB04b] and the Waters' hash [Wat05], reducing the $\mathsf{mpk}$ size to $\omega(\log^2 \lambda)$ and $\omega(\log \lambda)$, respectively. However, the modification of the Waters' hash requires a fixed but implicit modulus $q$ due to the use the Barrington's Theorem [Bar89]. For this reason, we only focus on the former case, i.e., the modified admissible hash as the partitioning function. To further reduce the modulus, Katsumata [Kat17] improved upon the partitioning function from [Yam17], achieving a lower-degree version. Below, we recall Katsumata's refined partitioning function.

**Definition 6 ([Yam17, Kat17]).** *Let* $\eta, \vartheta, \xi$ *be positive integers such that* $\eta = \omega(\log \lambda)$*,* $\vartheta = O(\log \lambda)$ *and* $\xi = O(\lambda)$*. Let* $\ell$ *be the length of the identity such that* $\ell = O(n)$*. Let* $C(\cdot)$ *be a public function that maps an identity to a bit string in* $\{0,1\}^\xi$*,* $S(\cdot) := \{2i - C(\cdot)_i | i \in [\xi]\}$ *where* $C(\cdot)_i$ *represents the* $i$*-th bit of* $C(\cdot)$*. The equality test function is defined as* $\mathsf{Equal}(v, u) = 1 - (v-u)^2$ *for some* $v, u \in \{0,1\}$*. Let the key space* $\mathcal{K} := \{0,1\}^{\eta\vartheta}$*, the identity space* $\mathcal{ID} := \{0,1\}^\ell$*, the range* $\mathcal{T} := \{0, 1, \cdots, \eta\}$*. Then, for a key* $\boldsymbol{\kappa} := \{\kappa_{i,j}\}_{(i,j)\in[\eta]\times[\vartheta]} \in \mathcal{K}$*, an identity* $\mathsf{id} \in \mathcal{ID}$*, the function* $H$ *is defined as*

$$H\left(\boldsymbol{\kappa} := \{\kappa_{i,j}\}_{(i,j)\in[\eta]\times[\vartheta]}, \mathsf{id}\right) := \eta - \sum_{i\in[\eta]} \sum_{i'\in[\xi]} \underbrace{\prod_{j\in[\vartheta]} \mathsf{Equal}(\kappa_{i,j}, s_{i',j})}_{\kappa_i \stackrel{?}{=} s_{i'}},$$

*where* $S(\mathsf{id}) = (s_1, \cdots, s_\xi)$ *and* $s_{i',j}$ *represent the* $j$*-th bit of* $s_{i'}$*.*

The following lemma characterizes important properties of the function $H$.

**Lemma 12 ([Yam17, Kat17]).** *The function $H$ defined above is a partitioning function (Def. 3). More specifically, there exists an algorithm $\mathsf{PrtSmp}_{\mathsf{Yam}}$ satisfying the requirement of [Def. 3, Item 1], and the quality of the partitioning function is $\tau := O\left(\frac{\epsilon^{\nu+1}}{Q^{\nu}}\right)$ for some constant $\nu \in \mathbb{N}$ depending on the function $C(\cdot)$, which is noticeable as required in [Def. 3, Item 2]. Furthermore, there exists two deterministic algorithms $(\mathsf{PubEval}_{\mathsf{Yam}}, \mathsf{TrapEval}_{\mathsf{Yam}})$ which are $\Delta$-expanding for the function family $\{H\}$ (Def. 4), where $\Delta := \tilde{O}(n\eta\vartheta\ell)$.*

The performance of the Yamada IBE can be quantified as follows.

$$|\mathsf{mpk}|\# = \omega(\log^2 \lambda), \quad \textit{LWE param } \frac{1}{\alpha} = \tilde{O}(n^{5.5}), \quad \textit{reduction cost} = O\left(\frac{\epsilon^{\nu+1}}{Q^{\nu}}\right).$$

The following corollary quantifies the performance of IBE which combines the Yamada IBE and our new framework.

**Corollary 2.** *If we plug the partitioning function $H$ in Def. 6 and the corresponding algorithms $(\mathsf{PrtSmp}_{\mathsf{Yam}}, \mathsf{PubEval}_{\mathsf{Yam}}, \mathsf{TrapEval}_{\mathsf{Yam}})$ (in Lemma 12) into the framework in Sect. 6.1, the resulting IBE scheme satisfies that*

$$|\mathsf{mpk}|\# = \omega(\log^2 \lambda), \quad \text{LWE param } \frac{1}{\alpha} = \tilde{O}(n^{3}), \quad \text{reduction cost} = O\left(\frac{\epsilon^{\nu+1}}{Q^{\nu}}\right).$$

*Proof.* This result follows directly from Theorem 4 and Lemma 12. □

### 7.3 Application to the ALWW IBE

Here, we apply our framework to the ALWW IBE [ALWW21]. Specifically, Abla et al. [ALWW21] proposed a new equality test function using algebraic structures of the rings, resulting in only $\omega(1)$ basic ring vectors in the public parameter.

**Definition 7 ([ALWW21]).** *Let $q$ be a prime modulus such that $q \equiv 3 \bmod 8$. Let $\mathcal{R} := \mathbb{Z}[x]/(x^n+1)$ be some cyclotomic ring with degree $n$ be a power of 2 and $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$. Let $t, p$ be positive integers such that $t \cdot p \leq n$ and $t = \omega(1)$. Let $L$ be a positive integer such that $L = O(n^{1+\frac{2}{\varrho}})$ for a constant $\varrho > 1$ that satisfies $n^{\frac{1}{\varrho}} > 3 + \varrho$. Let $\eta$ be a constant such that $L + 1 \leq (2n)^{\eta}$. Let $\ell$ be the length of the identity such that $\ell = O(n)$. Let $F$ be a public error correcting code that maps an identity to a vector in $\mathbb{Z}_p^L$ and $F(\mathsf{id})[0] = 0$ for any $\mathsf{id} \in \mathcal{ID}$. The equality test function is defined as $\mathsf{Equal}(u, v) = \frac{1}{2n} \cdot \sum_{i=0}^{2n-1} (x^{u-v})^i$ for some $u, v \in [2n]$. Let the key space $\mathcal{K} := [L]^t \times [p]^t$, the identity space $\mathcal{ID} := \{0,1\}^{\ell}$, and the range $\mathcal{T} := \mathcal{R}_q$. Then, for a key $\boldsymbol{\kappa} := (\boldsymbol{\alpha}, \boldsymbol{\beta}) \in \mathcal{K}$ where $\boldsymbol{\alpha} \in [L]^t$ and $\boldsymbol{\beta} \in [p]^t$, an identity $\mathsf{id} \in \mathcal{ID}$, the function $H$ is defined as*

$$\begin{aligned} &H\left(\boldsymbol{\kappa} := (\{\alpha_{i,j}\}_{(i,j)\in[t]\times[\eta]}, \boldsymbol{\beta}), \mathsf{id}\right) \\ &:= -\sum_{i=1}^{t} x^{ip+\beta_i} + \sum_{i=1}^{t} \sum_{z=1}^{L+1} \underbrace{\left(\prod_{j=1}^{\eta} \mathsf{Equal}\left(\alpha_{i,j}, z_j\right)\right)}_{z \stackrel{?}{=} \alpha_i} x^{ip+F(\mathsf{id})[z]}, \end{aligned}$$

*where $\alpha_i$ ($\beta_i$) is the $i$-th element of $\boldsymbol{\alpha}$ ($\boldsymbol{\beta}$), $\alpha_{i,j}$ is the $j$-th element in $\alpha_i$'s $2n$-ary representation $(\alpha_{i,1}, \cdots, \alpha_{i,\eta})$, and $z_j$ is the $j$-th element in $z$'s $2n$-ary representation $(z_1, \cdots, z_\eta)$.*

The following lemma characterizes important properties of the function $H$.

**Lemma 13 ([ALWW21]).** *The function $H$ defined above is a partitioning function (Def. 3). More specifically, there exists an algorithm $\mathsf{PrtSmp}_{\mathsf{ALWW}}$ satisfying the requirement of [Def. 3, Item 1], and the quality of the partitioning function is $\tau := O\left(\frac{\epsilon^2}{Q}\right)$ which is noticeable as required in [Def. 3, Item 2]. Furthermore, there exists two deterministic algorithms $(\mathsf{PubEval}_{\mathsf{ALWW}}, \mathsf{TrapEval}_{\mathsf{ALWW}})$ which are $\Delta$-expanding for the function family $\{H\}$ (Def. 4), where $\Delta := \tilde{O}(tL\eta n^3)$.*

The performance of the ALWW IBE can be quantified as follows.

$$|\mathsf{mpk}|\# = \omega(1), \quad \textit{LWE param } \frac{1}{\alpha} = \tilde{O}(n^{9.5+\frac{4}{\varrho}}), \quad \textit{reduction cost} = O\left(\frac{\epsilon^2}{Q}\right).$$

The following corollary quantifies the performance of IBE which combines the ALWW IBE and our new framework.

**Corollary 3.** *If we plug the partitioning function $H$ in Def. 7 and the corresponding algorithms $(\mathsf{PrtSmp}_{\mathsf{ALWW}}, \mathsf{PubEval}_{\mathsf{ALWW}}, \mathsf{TrapEval}_{\mathsf{ALWW}})$ (in Lemma 13) into the framework in Sect. 6.1, the resulting IBE scheme satisfies that*

$$|\mathsf{mpk}|\# = \omega(1), \quad \text{LWE param } \frac{1}{\alpha} = \tilde{O}(n^{5+\frac{2}{\varrho}}), \quad \text{reduction cost} = O\left(\frac{\epsilon^2}{Q}\right).$$

*Proof.* This result follows directly from Theorem 4 and Lemma 13. □

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**Table 2.** Preformance comparison of lattice IBE constructions under three frameworks: (1) original framework, (2) Ji et al.'s framework, (3) our framework.

| Scheme \ Performance (Framework) | $\lvert\mathsf{mpk}\rvert$ # of $\mathbb{Z}_q^{n\times m}$ mat. | LWE Param $\frac{1}{\alpha} = \frac{q}{\sigma_{\mathsf{LWE}}}$ | | | $\lvert\mathsf{sk_{id}}\rvert$, $\lvert\mathsf{ct}\rvert$ # of $\mathbb{Z}_q^m$ vec. | Reduction cost |
|---|---|---|---|---|---|---|
| | (1)/(2)/(3) | (1) Original | (2) Ji et al.'s[‡] | (3) Ours | (1)/(2)/(3) | (1)/(2)/(3) |
| [ABB10] + [Boy10][♯] | $O(\lambda)$ | $\tilde{O}(n^{3.5})$ | $\tilde{O}(n^4)$ | $\tilde{O}(n^2)$ | $O(1)$ | $O(\frac{\epsilon^3}{Q^2})$ |
| [ZCZ16][⋆] | $O(\log Q)$ | $\tilde{O}(Q^2 \cdot n^{6.5})$ | $\tilde{O}(Q \cdot n^{5.5})$ | $\tilde{O}(Q \cdot n^{3.5})$ | $O(1)$ | $O(\frac{\epsilon}{\ell Q^2})$ |
| [KY16][♮] | $O(\lambda^{1/d})$ | $\tilde{O}(n^{2d+1.5})$ | $\tilde{O}(n^{d+3})$ | $\tilde{O}(n^{d+1})$ | $O(1)$ | $O\left(\frac{\lambda^{d-1}\epsilon^d}{Q^d}\right)^{d+1}$ |
| [Yam17] I + [Kat17][*] | $\omega(\log^2 \lambda)$ | $\tilde{O}(n^{5.5})$ | $\tilde{O}(n^5)$ | $\tilde{O}(n^3)$ | $O(1)$ | $O(\frac{\epsilon^{\nu+1}}{Q^{\nu}})^{\bullet}$ |
| [LLW20][♭] | $O(\lambda)$ | $\tilde{O}(n^{5.5})$ | $\tilde{O}(n^5)$ | $\tilde{O}(n^3)$ | $O(1)$ | $O(\frac{\epsilon}{\lambda})$ |
| [JKN21] | $O(\log \lambda)$ | $\tilde{O}(n^{5.5})$ | $\tilde{O}(n^5)$ | $\tilde{O}(n^3)$ | $O(1)$ | $O(\frac{\epsilon}{t^2})$ |
| [ALWW21][†] | $\omega(1)$ | $\tilde{O}(n^{9.5+\frac{4}{\varrho}})$ | $\tilde{O}(n^{7+\frac{2}{\varrho}})$ | $\tilde{O}(n^{5+\frac{2}{\varrho}})$ | $O(1)$ | $O(\frac{\epsilon^2}{Q})$ |
| [Abl24][∘] | $\omega\left(\frac{\log\lambda}{\log\log\lambda}\right)$ | $\tilde{O}(n^{9.5})$ | $\tilde{O}(n^7)$ | $\tilde{O}(n^5)$ | $O(1)$ | $O(\frac{\epsilon^2}{Q})$ |

Notations: $\mathsf{mpk}$, $\mathsf{ct}$, and $\mathsf{sk_{id}}$ denote the master public key, ciphertext, and secret key of the IBE, respectively. $\epsilon$ and $t$ denote the advantage and the running time in attacking the IBE scheme. $\ell$ denotes the bit length of the identity. All the other notations are consistent with those in Tab. 1.

In this table, the parameter marked "(1)/(2)/(3)" achieves asymptotically the **same complexity** in the three frameworks (original, Ji et al.'s, and ours). We note that only the LWE parameters differ across the three frameworks for lattice IBE schemes.

[•] $\nu > 1$ is a constant that can be set small, depending on the underlying error correcting code.

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# A The Solution of [JWLG25]

Concurrently, Ji et al. [JWLG25] identified the same quadratic restriction and gave a different solution. In this section, we recall their solution and provide a detailed comparison with ours.

## A.1 Recall the solution of [JWLG25]

At a high level, their idea can be represented in Fig. 2.

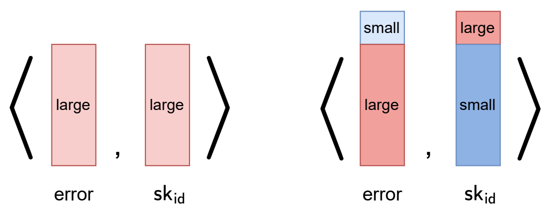


**Fig. 2.** The error term of origianl framework (left) and [JWLG25] (right), where $\langle\cdot,\cdot\rangle$ represents vector inner product. The color depth represents the relative Gaussian width of the components, where the deeper the color, the larger the Gaussian width.

Specifically, their goal is to obtain a $(D_{\mathbb{Z}^n,\theta_1}, D_{\mathbb{Z}^{2m},\theta_2})$-hybrid secret key where $\theta_1 \gg \theta_2$, and a $(D_{\mathbb{Z}^n,\sigma_1}, D_{\mathbb{Z}^{2m},\sigma_2})$-hybrid error where $\sigma_1 \ll \sigma_2$. Note that only the big ones $\theta_1$ and $\sigma_2$ are *linear* in $\|\mathbf{R}_{\mathsf{id}}\|$, but the small ones $\theta_2$ and $\sigma_1$ are not.

*Obtain a $(D_{\mathbb{Z}^n,\theta_1}, D_{\mathbb{Z}^{2m},\theta_2})$-hybrid secret key.* Their $\mathsf{mpk}$ consists of $t$ GSW-style encodings[7] and 1 BGG+-style encoding:

$$\mathsf{mpk} := \left(\mathbf{B},\ \mathbf{C} := \mathbf{B}\mathbf{R}_s + \mathbf{S}\mathbf{G},\ \mathbf{C}_i := \left\{\begin{bmatrix}\mathbf{A}\\ \mathbf{S}\mathbf{A}+\mathbf{E}\end{bmatrix}\mathbf{R}_i + \kappa_i\begin{bmatrix}\mathbf{G} & \\ & \mathbf{G}\end{bmatrix}\right\}_{i\in[t]},\ \mathbf{D}\right), \tag{8}$$

where $(\mathbf{A}, \mathbf{S}\mathbf{A}+\mathbf{E}) \in \mathbb{Z}_q^{n\times m} \times \mathbb{Z}_q^{n\times m}$ are some multiple-secret LWE samples, $\begin{bmatrix}\mathbf{G} & \\ & \mathbf{G}\end{bmatrix} \in \mathbb{Z}_q^{2n\times 2m}$ is a block diagonal matrix, $\{\mathbf{R}_i\}_{i\in[t]} \in \mathbb{Z}^{m\times 2m}$, $\mathbf{R}_s \in \mathbb{Z}^{m\times m}$ are some random matrices with small entries, and $\mathbf{D} \in \mathbb{Z}_q^{n\times n}$ is an invertible matrix over $\mathbb{Z}_q$. They compute the user's public key $\mathsf{pk}_{\mathsf{id}}$ in three steps:

1. Let the $t$ GSW-style encodings $\left\{\mathbf{C}_i := \begin{bmatrix}\mathbf{A}\\ \mathbf{S}\mathbf{A}+\mathbf{E}\end{bmatrix}\mathbf{R}_i + \kappa_i\begin{bmatrix}\mathbf{G} & \\ & \mathbf{G}\end{bmatrix}\right\}_{i\in[t]}$ and $\mathsf{id}$ be the inputs, there exists an algorithm that homomorphically computes the partitioning function $H(\cdot,\cdot)$ and outputs $\begin{bmatrix}\mathbf{A}\\ \mathbf{S}\mathbf{A}+\mathbf{E}\end{bmatrix}\mathbf{R}_{\mathsf{id}} + H(\boldsymbol{\kappa},\mathsf{id})\begin{bmatrix}\mathbf{G} & \\ & \mathbf{G}\end{bmatrix}$.
2. Take the last $m$ columns of the output in the first step, i.e., $\widehat{\mathbf{C}} = \begin{bmatrix}\widehat{\mathbf{C}}_0\\ \widehat{\mathbf{C}}_1\end{bmatrix} = \begin{bmatrix}\mathbf{A}\\ \mathbf{S}\mathbf{A}+\mathbf{E}\end{bmatrix}\widehat{\mathbf{R}_{\mathsf{id}}} + H(\boldsymbol{\kappa},\mathsf{id})\begin{bmatrix}\mathbf{0}\\ \mathbf{G}\end{bmatrix}$, where $\widehat{\mathbf{R}_{\mathsf{id}}}$ is the last $m$ columns of $\mathbf{R}_{\mathsf{id}}$.

[7] Gentry et al. [GSW13] showed that the GSW-style encoding not only supports homomorphic addition and multiplication operations, but can also be decoded with the help of the LWE secret $\mathbf{S}$.

3. Let the BGG+-style encoding $\mathbf{C} := \mathbf{B}\mathbf{R}_s + \mathbf{S}\mathbf{G}$ and $\widehat{\mathbf{C}}$ be the inputs, compute an "incomplete decryption" function as follows

$$\widehat{\mathbf{C}}_1 - \mathbf{C} \cdot \mathbf{G}^{-1}(\widehat{\mathbf{C}}_0) = \mathbf{B} \cdot \underbrace{(-\mathbf{R}_s \cdot \mathbf{G}^{-1}(\widehat{\mathbf{C}}_0))}_{:=\mathbf{M}_{\mathsf{id}}} + H(\boldsymbol{\kappa}, \mathsf{id})\mathbf{G} + \underbrace{\mathbf{E} \cdot \widehat{\mathbf{R}_{\mathsf{id}}}}_{:=\mathbf{E}_{\mathsf{id}}}. \tag{9}$$

Now the user's public key $\mathsf{pk}_{\mathsf{id}}$ can be written as

$$\mathsf{pk}_{\mathsf{id}} := \mathbf{D} \cdot [\mathbf{I}_n | \mathbf{B} | \mathbf{B}\mathbf{M}_{\mathsf{id}} + H(\boldsymbol{\kappa}, \mathsf{id})\mathbf{G} + \mathbf{E}_{\mathsf{id}}] \in \mathbb{Z}_q^{n \times (n+2m)}. \tag{10}$$

For a given uniform vector $\mathbf{u} \in \mathbb{Z}_q^n$, the user's secret key $\mathsf{sk}_{\mathsf{id}}$, a $n+2m$-length vector satisfying $\mathsf{pk}_{\mathsf{id}} \cdot \mathsf{sk}_{\mathsf{id}} = \mathbf{u}$, can be sampled with the secret trapdoors $\mathbf{M}_{\mathsf{id}}$, $\mathbf{E}_{\mathsf{id}}$ and the public trapdoor $\mathbf{T}_{\mathbf{G}}$. More specifically, $\mathsf{sk}_{\mathsf{id}}$ consists of two parts: a $n$-length vector with large entries and a $2m$-length vector with small entries, i.e., $\mathsf{sk}_{\mathsf{id}} := \left[\begin{smallmatrix}\mathbf{x}_1 \\ \mathbf{x}_2\end{smallmatrix}\right] \in \mathbb{Z}^{n+2m}$, and satisfies the following equation

$$[\mathbf{B} | \mathbf{B}\mathbf{M}_{\mathsf{id}} + H(\boldsymbol{\kappa}, \mathsf{id})\mathbf{G} + \mathbf{E}_{\mathsf{id}}] \cdot \underbrace{\left(\mathbf{p}_1 + \begin{bmatrix} -\mathbf{M}_{\mathsf{id}} \\ \mathbf{I} \end{bmatrix} \mathbf{z}\right)}_{\mathbf{x}_2 := \text{ small part of } \mathsf{sk}_{\mathsf{id}}} = \mathbf{D}^{-1}\mathbf{u} + \underbrace{(\mathbf{p}_2 + \mathbf{E}_{\mathsf{id}}(\mathbf{h} + \mathbf{z}))}_{-\mathbf{x}_1 := \text{ large part of } \mathsf{sk}_{\mathsf{id}}}. \tag{11}$$

where $\mathbf{p}_1 \in \mathbb{Z}^{2m}, \mathbf{p}_2 \in \mathbb{Z}^m, \mathbf{h} \in \mathbb{Z}^m$ are perturbations to ensure that no information about the trapdoor matrices $\mathbf{M}_{\mathsf{id}}$, $\mathbf{E}_{\mathsf{id}}$ and the secret vector $\mathbf{z}$ are revealed, and $\mathbf{z} \in \mathbb{Z}^m$ follows a discrete Gaussian distribution such that $\mathbf{G}\mathbf{z} = H(\boldsymbol{\kappa}, \mathsf{id})^{-1} \cdot \big((\mathbf{D}^{-1}\mathbf{u} + \mathbf{p}_2 + \mathbf{E}_{\mathsf{id}}\mathbf{h}) - [\mathbf{B} | \mathbf{B}\mathbf{M}_{\mathsf{id}} + H(\boldsymbol{\kappa}, \mathsf{id})\mathbf{G} + \mathbf{E}_{\mathsf{id}}] \cdot \mathbf{p}_1\big)$. The Gaussian convolution technique [Pei10] ensures that $\mathsf{sk}_{\mathsf{id}}$ is distributed statistically close a non-spherical Gaussian $D_{\mathbb{Z}^{n+2m}, \sqrt{\Sigma}}$ where $\Sigma := \left[\begin{smallmatrix}\theta_1^2 \mathbf{I}_n & \\ & \theta_2^2 \mathbf{I}_{2m}\end{smallmatrix}\right]$, under the condition that $\theta_1 = \tilde{O}(\zeta \cdot \|\mathbf{E}_{\mathsf{id}}\|)$, $\theta_2 = \tilde{O}(\zeta \cdot \|\mathbf{M}_{\mathsf{id}}\|)$ and $\zeta$ is Gaussian width of $\mathbf{z}$.

*Obtain a $(D_{\mathbb{Z}^n, \sigma_1}, D_{\mathbb{Z}^{2m}, \sigma_2})$-hybrid error.* They set the ciphertext with a hybrid error as

$$\mathsf{ct} := \left(c_0 := \mathbf{v}^\top \mathbf{u} + e_0 + \left\lceil \frac{q}{2} \right\rceil \cdot \mu, \quad \mathbf{c}^\top := \mathbf{v}^\top \mathsf{pk}_{\mathsf{id}} + [\mathbf{e}_1^\top | \mathbf{e}_2^\top]\right) \in \mathbb{Z}_q \times \mathbb{Z}_q^{n+2m}, \tag{12}$$

where $\mathbf{v} \xleftarrow{\$} \mathbb{Z}_q^n$ and $(e_0, \mathbf{e}_1, \mathbf{e}_2) \leftarrow D_{\mathbb{Z}, \delta} \times D_{\mathbb{Z}^n, \sigma_1} \times D_{\mathbb{Z}^{2m}, \sigma_2}$[8]. The public key for the challenge identity $\mathsf{id}^*$ can be written as $\mathsf{pk}_{\mathsf{id}^*} := \mathbf{D} \cdot [\mathbf{I}_n | \mathbf{B} | \mathbf{B}\mathbf{M}_{\mathsf{id}^*} + \mathbf{E}_{\mathsf{id}^*}]$ since $H(\boldsymbol{\kappa}, \mathsf{id}^*) = 0$. Thus, the $(n+2m)$-length vector $\mathbf{c}^*$ in the challenge ciphertext can be simulated as follows:

$$\begin{aligned}
\mathbf{c}^* &:= \begin{bmatrix} \mathbf{D}^\top \mathbf{v} + \mathbf{e}_1 \\ \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{M}_{\mathsf{id}^*}^\top & \mathbf{E}_{\mathsf{id}^*}^\top \end{bmatrix} \left( \begin{bmatrix} \mathbf{B}^\top \mathbf{D}^\top \\ \mathbf{D}^\top \end{bmatrix} \mathbf{v} + \mathbf{e}_1' \right) + \mathbf{p} \end{bmatrix} \\
&= \begin{bmatrix} \mathbf{I} \\ \mathbf{B}^\top \\ \mathbf{M}_{\mathsf{id}^*}^\top \mathbf{B}^\top + \mathbf{E}_{\mathsf{id}^*}^\top \end{bmatrix} \mathbf{D}^\top \mathbf{v} + \begin{bmatrix} \mathbf{e}_1 \\ \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{M}_{\mathsf{id}^*}^\top & \mathbf{E}_{\mathsf{id}^*}^\top \end{bmatrix} \mathbf{e}_1' + \mathbf{p} \end{bmatrix} = \mathsf{pk}_{\mathsf{id}^*}^\top \mathbf{v} + \begin{bmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \end{bmatrix},
\end{aligned}$$

[8] In their setting, $\delta = \sigma_1$.

where $\left(\begin{bmatrix}\mathbf{B}^\top\mathbf{D}^\top\\ \mathbf{D}^\top\end{bmatrix}, \begin{bmatrix}\mathbf{B}^\top\mathbf{D}^\top\\ \mathbf{D}^\top\end{bmatrix}\mathbf{v}+\mathbf{e}_1'\right)$ are some LWE samples and $\mathbf{p}\in\mathbb{Z}^{2m}$ is a perturbation to ensure that no information about the trapdoor matrices $\mathbf{M}_{\mathsf{id}^*}$, $\mathbf{E}_{\mathsf{id}^*}$ are revealed. The re-randomization technique guarantees that the simulated $\mathbf{e}_2 := \begin{bmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{M}_{\mathsf{id}^*}^\top & \mathbf{E}_{\mathsf{id}^*}^\top\end{bmatrix}\mathbf{e}_1' + \mathbf{p}$ is distributed statistically close to $D_{\mathbb{Z}^{2m},\sigma_2}$ under the condition that $\sigma_1 = \tilde{O}(\delta)$, $\sigma_2 = \tilde{O}\left(\delta\cdot\left\|\begin{bmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{M}_{\mathsf{id}^*}^\top & \mathbf{E}_{\mathsf{id}^*}^\top\end{bmatrix}\right\|\right)$, and $\delta$ is the Gaussian width of the LWE error $\mathbf{e}_1, \mathbf{e}_1'$.

Therefore, they obtain a $(D_{\mathbb{Z}^n,\theta_1}, D_{\mathbb{Z}^{2m},\theta_2})$-hybrid secret key where $\theta_1 \gg \theta_2$, and a $(D_{\mathbb{Z}^n,\sigma_1}, D_{\mathbb{Z}^{2m},\sigma_2})$-hybrid error where $\sigma_1 \ll \sigma_2$. During the decryption step, the user who owns the secret key $\mathsf{sk}_{\mathsf{id}}$ can compute

$$c_0 - \mathbf{c}^\top\cdot\mathsf{sk}_{\mathsf{id}} = \left\lceil\frac{q}{2}\right\rceil\cdot\mu + \underbrace{e_0 - \left\langle\begin{bmatrix}\mathbf{e}_1\\ \mathbf{e}_2\end{bmatrix}, \begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix}\right\rangle}_{:=\text{error term}}. \tag{13}$$

In summary, they achieved cross multiplication in the error term as shown in Fig. 2, and thus removed the quadratic restriction.

### A.2 Comparison with ours

In this subsection, we compare Ji et al.'s framework and ours in three aspects: (1) cross-multiplication structure; (2) IBE construction, and (3) IBE performance.

*Cross-multiplication structure.* Even though the cross-multiplication structure is used in both solutions, our approach directly uses the **natural** cross-multiplication structure in the original framework, whereas Ji et al. generate an **artificial** cross-multiplication structure by extensive modification to the original framework.

A comparison between the cross-multiplication structures of our approach and theirs is shown in Fig. 3.

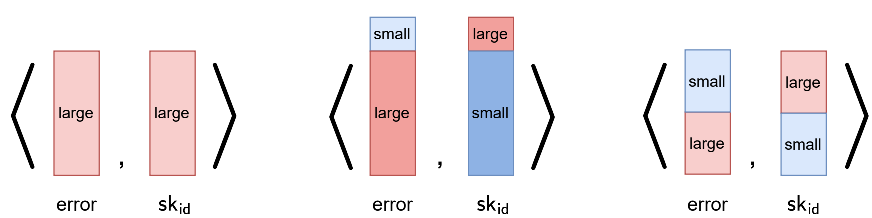


**Fig. 3.** The error term of origianl framework (left), [JWLG25] (middle) and ours (right). The color depth represents the relative Gaussian width of the components, where the deeper the color, the larger the Gaussian width.

*IBE construction.* Recall our construction in Sect. 2 and Ji et al.'s construction in Sect. A.1, we can find that Ji et al.'s framework is far more cumbersome than ours. In more detail, they require extensive modification to the original framework that include:

1. public matrices in two different encoding forms, i.e., BGG+-style encoding $\mathbf{C}$ and GSW-style encodings $\{\mathbf{C}_i\}_{i\in[t]}$ in Eq. (8)),
2. extra homomorphic computation for an "incomplete decryption" function as in Eq. (9),
3. more complex dsigns of the user's public key (Eq. (10)), secret key (Eq. (11)) and the ciphertexts (Eq. (12)).

In contrast, our approach does not introduce any additional steps to the original framework, nor structural modifications to any components. The only changes lie in the Gaussian widths of the secret key and error as well as the simulation of such non-spherical distributions in the security proof. Thus, our framework adapts the original framework in a more straightforward way and is significantly simpler than Ji et al.'s framework.

*IBE performance.* Compared to Ji et al.'s solution, we provide a better-performing solution. More specifically, our solution has the following improvements in performance:

1. *Smaller modulus.* Our modulus is smaller than theirs for two reasons.
   - *Smaller dimensions.* Their construction extends the secret key (Eq. (11)) and the error (Eq. (12)) dimensions from $2m$ to $2m+n$, while our framework keeps the dimensions unchanged.
   - *Smaller Gaussian widths.* Their method introduces extra terms that amplify the Gaussian widths of both secret key and errors, resulting in an extra overhead of overall $\tilde{O}(n^2)$ compared to our modulus. More precisely,
     - As shown in Sect. A.1, the Gaussian width of their secret key is $\theta_2 = \tilde{O}(\zeta \cdot \|\mathbf{M}_{\mathsf{id}}\|)$, the Gaussian width of their error is $\sigma_2 = \tilde{O}(\delta \cdot \|\mathbf{E}_{\mathsf{id}}\|)$, thus their modulus[9] is $q = \tilde{O}(\theta_2 \cdot \sigma_2 \cdot \sqrt{m}) = \tilde{O}(\zeta\delta \cdot \|\mathbf{M}_{\mathsf{id}}\| \cdot \|\mathbf{E}_{\mathsf{id}}\| \cdot \sqrt{m})$. By substituting $\mathbf{M}_{\mathsf{id}} \coloneqq -\mathbf{R}_s \cdot \mathbf{G}^{-1}(\cdot)$ and $\mathbf{E}_{\mathsf{id}} \coloneqq \mathbf{E} \cdot \widehat{\mathbf{R}_{\mathsf{id}}}$ as in Eq. (9), their modulus becomes $q = \tilde{O}(\zeta\delta \cdot \|\mathbf{R}_s\| \cdot \|\mathbf{G}^{-1}(\cdot)\| \cdot \|\mathbf{E}\| \cdot \|\mathbf{R}_{\mathsf{id}}\| \cdot \sqrt{m})$.
     - In contrast, our approach does not introduce any extra terms into the modulus. More precisely, the Gaussian width of our secret key is $\theta_2 = \tilde{O}(\zeta)$ and the Gaussian width of our error $\sigma_2 = \tilde{O}(\delta \cdot \|\mathbf{R}_{\mathsf{id}}\|)$, thus our modulus is $q = \tilde{O}(\zeta\delta \cdot \|\mathbf{R}_{\mathsf{id}}\| \cdot \sqrt{m})$.

   Compared with ours, their approach introduces extra terms $\mathbf{E}$, $\mathbf{R}_s$, $\mathbf{G}^{-1}(\cdot)$ to the modulus, resulting in an extra overhead of overall $\tilde{O}(n^2)$.

[9] Here, we don't need to consider the parameters $\theta_1$ and $\sigma_1$ since $\tilde{O}(\theta_1 \cdot \sigma_1) \leq \tilde{O}(\theta_2 \cdot \sigma_2)$.

2. *Smaller ciphertext size.* Our ciphertext $\big(\text{in } \mathbb{Z}_q \times \mathbb{Z}_q^{2m}\big)$ size is smaller than theirs $\big(\text{in } \mathbb{Z}_q \times \mathbb{Z}_q^{2m+n}\big)$ for two reasons.
   - *Smaller dimension.* Their construction extends the second part of the ciphertext (Eq. (12)) from $2m$ to $2m+n$, while our framework remains the same.
   - *Smaller modulus.* As described above, our modulus is smaller than theirs.
3. *More compact master public key.* Our master public key $\big(\text{in } \{\mathbb{Z}_q^{n\times m}\}_{i\in[t]}\big)$ is more compact than theirs $\big(\text{in } \{\mathbb{Z}_q^{2n\times 2m}\}_{i\in[t]}\big)$ for two reasons.
   - *Smaller dimensions.* They use the GSW-style encodings in the master public key (Eq. (8)), whose size is $\mathbb{Z}_q^{2n\times 2m}$. In contrast, we use the BGG+-style encoding and keep the size $\mathbb{Z}_q^{n\times m}$ unchanged.
   - *Smaller modulus.* As described above, our modulus is smaller than theirs.

Meanwhile, their approach offers a unique advantage over both the original framework and ours: the ability to achieve the smallest secret key size. This is because only the small part of the secret key needs to be stored—the large part can be computed from it, as demonstrated in [JWLG25, Remark 1].